

Monetary Policy Transmission, Bank Market Power, and Income Source*

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June 22, 2026

Abstract

We study the role of bank non-interest income in deposit pricing and monetary policy transmission. Empirical evidence shows that banks with high deposit-related non-interest income charge higher fees, set higher deposit spreads, and exhibit lower deposit rate pass-through of monetary policy shocks. We develop and calibrate a banking model in which deposit rates and fees govern the intensive and extensive margins of deposit demand, respectively. Counterfactual analyses show that the relationship between non-interest income and deposit pricing is primarily driven by underlying differences in deposit market power, and that non-interest income has important implications for credit supply, bank risk, and deposit pricing.

JEL Classification: D43, E44, E52, G21, G51

Keywords: monetary policy, banks, pass-through, market power

*We are grateful for comments from Viral Acharya, Dean Corbae, Itamar Drechsler, Pablo D’Erasmus, Miguel Faria-e-Castro, Iftekhar Hasan, Ivan Ivanov, Anil Kashyap, Yiming Ma, Ayşegül Şahin, Jan-Peter Siedlarek (discussant), Philipp Schnabl, Wenhao Li, Laurent Weill, and seminar participants at the Federal Reserve Bank of Kansas City, T2M Amsterdam 2024, CEBRA Annual Meeting 2024, Midwest Macroeconomics Meetings Fall 2024, System Committee on Financial Institutions, Regulation, and Markets Conference 2024, SEA 94th Annual Meeting, Wisconsin-Queens IO-Finance Reading Group, 2025 SED-Copenhagen, and EEA-Congress 2025. We also thank Thomas Desire for excellent research assistance. Views and opinions expressed in this paper reflect those of the authors and do not necessarily reflect those of the Federal Reserve System.

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1 Introduction

A central question in monetary economics is how monetary policy is transmitted to consumers and firms through the banking system. While an extensive literature studies the pass-through of monetary policy to loan and deposit rates (e.g., Drechsler et al. 2017 and Kashyap and Stein 2000), little focus has been given to the role of non-interest income—an increasingly important source of bank revenue in recent decades. In the United States, for example, non-interest income accounts for nearly 30% of aggregate bank operating income.¹ A substantial portion of this income comes from fees associated with deposit accounts, suggesting that any analysis of deposit pricing must account for both interest and non-interest sources of bank income.

This paper examines the role of bank non-interest income in deposit pricing and the pass-through of monetary policy to deposit rates. We specifically focus on non-interest income generated directly from deposit accounts—which we term deposit non-interest income (deposit NII)—arising from account fees and liquidity services provided to depositors.²

We document that banks with higher deposit non-interest income earn higher prices and markups on deposit accounts. Using a comprehensive measure of deposit account prices that includes both deposit spreads and non-interest income, we find that high NII banks earn \$199 annually per deposit account, compared with \$170 for low NII banks. Combining these prices with bank cost data, we find substantially higher markups among high NII banks, reinforcing the view that high non-interest income banks exert pricing power. We further show that deposit NII is a large and stable source of bank income, exhibiting low cyclicalities and low volatility, and becomes particularly important during periods of low interest rates, when interest income is compressed.

Banks with higher deposit non-interest income also exhibit significantly lower deposit rate pass-through of monetary policy shocks. Using branch-level data and local projection methods à la Jordà (2005), we examine whether the response of bank rates to monetary policy shocks from Jarociński (2024) varies with banks’ deposit non-interest income. While loan rates are unaffected, we find significant heterogeneity in deposit rates. In response to a 100 basis point monetary shock, deposit rates at high NII banks increase by 20 basis points less than at low NII banks. These results are strongest for accounts associated with liquidity services, such as checking and savings accounts.

¹In the largest euro area countries, the share of non-interest income is similar, or higher, and has been increasing since the Great Financial Crisis (World Bank, 2025).

²Liquidity services comprise non-price characteristics such as branch networks, online banking, and related financial services that increase the value of a deposit account from the consumer’s perspective.

Importantly, the non-interest income channel we identify operates independently of traditional drivers of rate pass-through—such as market concentration, capitalization, or size—highlighting the importance of bank income composition for monetary policy transmission.

An illustrative model of two-part deposit pricing shows how differences in deposit market power can rationalize the empirical findings. In the model, deposit rates govern the intensive margin of deposit demand—how many deposits consumers choose to hold—while account fees govern the extensive margin, namely whether to open or close an account. Banks with greater deposit market power set higher fees and spreads, and exhibit lower monetary policy pass-through. Thus, variation in deposit market power is the key underlying force in the model that rationalizes the empirical findings. While we do not rule out other mechanisms, the illustrative model shows how market power alone can generate the observed relationship between bank income and deposit pricing.

Using insights from the illustrative model, we next develop and calibrate a quantitative model of the US banking industry to perform counterfactual analyses. The model incorporates several novel features, including bank non-interest income and separate extensive and intensive margins of deposit demand. We use the model to examine the implications of non-interest income for credit supply, bank risk, and deposit pricing.

In the first counterfactual, we find that exogenous increases in bank non-interest income reduce deposit-rate pass-through of monetary policy shocks by lowering banks’ reliance on deposit funding. While this result is consistent with the empirical correlations documented earlier, the quantitative effects are modest. As a result, the model implies that most of the variation in deposit-rate pass-through is driven by differences in market power and other market frictions rather than by non-interest income itself.

In the second counterfactual, we find that fee caps reduce credit supply and increase bank leverage, leading to higher bank risk. Motivated by recent policy proposals aimed at limiting bank fees, we use the model to study the consequences of imposing fee caps on banks.³ Fee caps reduce non-interest income and increase the likelihood that banks enter low networth states, in which their demand for leverage is high, thereby increasing bank risk as measured by z-scores. The result highlights a tension between consumer protection and financial stability: while fee caps lower the cost of financial access and services, they may also increase fragility in the banking system and generate broader economic costs.

³See, for example, the regulation on non-sufficient funds fees (CFPB, January 2024; Federal Register, January 2024) or credit card late fees (CFPB, March 2024; Federal Register, March 2024).

Last, we show that competition generally reduces bank risk, but the effect depends on the margin of competition and bank type. In the counterfactual, we model competition as an increase in banks’ relevant demand elasticities and distinguish between competition along the intensive and extensive margins of deposit demand. In general, competition makes deposit funding more expensive, inducing banks to deleverage, thereby reducing risk. However, this result is not universal. For example, when a low NII bank faces increased competition on the extensive margin, a leverage motive can dominate due to a cross-subsidizing feature of deposit account pricing, leading to higher leverage and risk. These findings suggest that the relationship between competition and bank risk is not monotonic (e.g., Martinez-Miera and Repullo 2010 and Whited et al. 2021) and that the margin through which competition operates is central.

Related Literature. Our research contributes to a large literature examining the effects of monetary policy on bank pricing and balance sheet outcomes (e.g., Kashyap and Stein 2000, Bruno and Shin 2015, Altavilla et al. 2020 and Jimenez et al. 2012).⁴ More recently, attention has shifted toward the role of bank and market characteristics—such as size and market concentration—in shaping, or in relation to, bank pricing power and monetary policy pass-through (e.g., Drechsler et al. 2017, d’Avernas et al. 2023, Gödl-Hanisch 2023, Wang et al. 2022, Xiao 2020, Scharfstein and Sunderam 2016, and Narayanan and Ratnadiwakara 2025). We contribute to this literature by examining whether variation in bank non-interest income, particularly income generated from deposit accounts, has meaningful implications for bank pricing and monetary policy transmission.

To conduct counterfactual analysis, we develop a quantitative dynamic bank model where banks have deposit market power. As such, we contribute to a growing class of related quantitative models (e.g., Corbae and D’Erasmus 2021, Dempsey 2026, Ulate 2021, Pancost and Robatto 2023, Morelli et al. 2024, and Abadi et al. 2023).⁵ We extend the prototypical loan-deposit banking model by incorporating two margins of deposit demand: an intensive margin governed by deposit rates and an extensive margin for account demand governed by account fees. As a result, banks earn both net interest income and non-interest income from holding deposits and issuing loans.

In large part, we interpret our empirical findings and use of certain model assumptions through the lens of findings from the bank industrial organization literature (e.g., Egan et al. 2017,

⁴On the deposit side, the literature documents stickiness in deposit rates and sluggish pass-through, particularly for increases in policy rates. Examples include Berger and Hannan (1989), Diebold and Sharpe (1990), Neumark and Sharpe (1992), Driscoll and Judson (2013), Berlin and Mester (2015), and Yankov (2023).

⁵See also Begenau and Landvoigt 2022, Bianchi and Bigio 2022, Faria-e Castro 2020, Gertler et al. 2020, and Jamilov and Monacelli (2026).

d’Avernas et al. 2023, Allen et al. 2019, Benetton et al. 2025, Haddad et al. 2023, Egan et al. 2025, and Buchak and Jorring 2026). In this literature, demand for bank deposits and loans depends not only on interest rates but also non-rate characteristics related to the quality of the banks’ products and services. Consistent with this framework, we take steps to empirically identify deposit market power and incorporate it into our modeling framework, while being agnostic about the underlying source of market power.

Our research also relates to an empirical literature examining the relationship between non-interest income and bank risk (e.g., Brunnermeier et al. 2020; DeYoung and Roland 2001; Lepetit et al. 2008; Stiroh 2004, 2006; Stiroh and Rumble 2006). This literature generally finds a positive correlation and characterizes non-interest income as a volatile source of revenue compared with net interest income. While this characterization may apply to nontraditional activities such as brokerage, insurance, and investment banking, we specifically focus on non-interest income generated from deposit accounts and related services, which we call deposit NII. In our application, we find a different result: deposit NII is a stable source of income for banks.

Relatedly, there is a large literature examining the relationship between bank competition and financial stability. Early work emphasized the competition-fragility view, whereby increased competition erodes franchise value and encourages banks to take on more risk (e.g., Keeley 1990, Allen and Gale 2004, and Corbae and Levine 2025). Some more recent work, however, has also documented channels through which greater competition may reduce fragility and bank risk, for example through improved loan performance (e.g., Boyd and De Nicolo 2005), while also emphasizing that the relationship between competition and stability need not be monotonic (e.g., Martinez-Miera and Repullo 2010 and Whited et al. 2021). Although much of this literature focuses on loan market power and is therefore not directly comparable to our setting, our results align most closely with the latter view: greater competition generally improves stability in our model, but the magnitude and direction of the effects vary substantially across banks and margin of competition.

The remainder of the paper is organized as follows. Section 2 defines deposit non-interest income and quantifies its relevance for deposit pricing and other bank characteristics. Section 3 presents the main empirical analysis and evidence on non-interest income and monetary policy transmission. Section 4 develops an illustrative model that highlights a market power mechanism that links non-interest income to deposit pricing. Section 5 develops a quantitative model of the US banking system and performs counterfactual analyses. Section 6 concludes the paper.

2 Bank Non-Interest Income

Data. The primary source for rates and fees is bank Call Reports as well as RateWatch, provided by S&P Global Market Intelligence since 2018. RateWatch regularly surveys nearly 100k financial institution locations and collects quotes of deposits, loan rates, and fees at the branch and product level. The set of rates contains several deposit products: savings, money market, interest checking, CDs of various maturities, and also conditions upon the size of accounts. We further merge the data with the Statistics on Depository Institutions (SOD) from the Federal Deposit Insurance Corporation (FDIC) to obtain information on bank branch networks. We also utilize bank balance sheet and income statements from the Call Reports to construct bank-level measures of deposit account pricing and other bank characteristics. We focus on a sample period between 2000 and 2024. Refer to Appendix Section E for information on data definitions and variable construction.

Deposit Non-Interest Income. While the majority of bank income is generated from interest-bearing assets, non-interest income makes up a significant component of total bank income. Table 1 shows that, in the aggregate, non-interest income accounts for almost one third of bank income (Column 1). Although banks generate non-interest income from many sources, we focus on the portion directly earned from deposit accounts, which we term *deposit non-interest income* or deposit NII. Deposit NII consists of fees charged on deposit accounts and related liquidity services, representing an additional margin through which banks can extract value. These fees include subscription fees, payment processing fees, penalty fees (e.g., from overdraft or minimum balance), and fees from miscellaneous activities. For the median bank-quarter (Column 2), deposit NII constitutes 45% of total non-interest income, although there is considerable variation across bank-quarter observations (Columns 3-4).

TABLE 1: BANK INCOME STATISTICS: 2000-2024

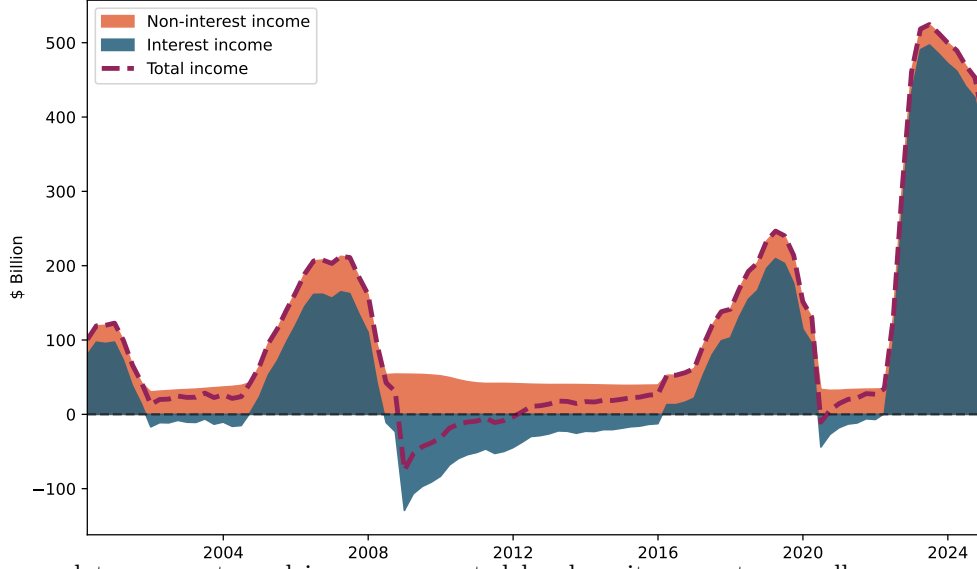
Moment	Aggregate	Bank-Quarter Observations		
	Sum	Median	10th Percentile	90th Percentile
Int Income to Operating Income	71.2	90	76.1	96.9
Non-Int Income (NII) to Operating Income	28.8	10	3.1	23.9
Deposit NII to NII	14.7	45	0	87.3
Non-traditional NII to NII	30.7	1.2	0	36.5
Asset NII to NII	11.8	0	-1.1	42
Other NII to NII	42.9	26.9	2.8	81

Notes: The first column (*Aggregate*) corresponds to a simple summation over all bank-quarters. The last three columns correspond to percentiles in the distribution of bank-quarter observations. Deposit NII includes fee income generated from liquidity services provided to owners of deposit accounts and includes payment services, branch services, mobile banking, and penalty fees. Non-traditional NII includes income from brokerage, insurance, proprietary trading, and investment banking. Asset NII includes loan servicing, securitization, loan origination fees, and asset value gains/losses. Other NII is a residual category. Refer to Section E for data definitions. Source: Call Report.

Figure 1 decomposes aggregate deposit income into interest and non-interest components, and emphasizes the importance of deposit non-interest income, particularly during periods of low interest rates (e.g., 2002-2004, 2008-2016, and 2020-2022). We construct the interest income measure using bank-level deposit spreads—i.e., the federal funds rate minus the savings rate—scaled by the quantity of deposits, thereby capturing the foregone interest paid by depositors when deposit rates are below market rates.⁶ During periods of low interest rates, banks often pay deposit rates above market rates, implying that deposit interest income turns negative. Thus, while deposit NII represents a small share of total income, on average, the figure suggests it is a stable income source that plays a larger role during periods of low interest rates. Similarly, bank deposit fees are stable over time and invariant to monetary policy (Appendix Figure A.1). At the branch level, monthly fees for savings or money market accounts, as well as fees for transactions such as bill pay, out-of-network and cashier’s checks, and fees for incoming and outgoing wire transfers, exhibit a similar pattern (Appendix Figure A.2).

⁶We exclude loan interest to focus solely on income generated from deposit accounts. Thus, one can think of the interest income generated by the deposit spread as the income of a custodian-type bank which borrows at low rates and invests in liquid, short-term assets. Of course, deposit interest income is often well-hedged by asset interest income (e.g., see Drechsler et al. 2021), so Figure 1 provides an incomplete picture of total bank profitability.

FIGURE 1: AGGREGATE DEPOSIT ACCOUNT INCOME



Notes: The figure plots aggregate real income generated by deposit accounts as well as components related to interest income and non-interest income. *Interest income* is calculated at the bank level with the product of deposit spreads and quantity of total deposits, and then is summed across banks. All income components are measured in real terms, using the PCE price deflator. Refer to Appendix Section E for explicit definitions. Source: Call Report.

Compared to other bank income categories, deposit non-interest income is relatively stable. Table 2 reports the growth volatility of aggregate bank income components. Separately, interest income and expense are volatile (5.2% and 20.9%, respectively), although hedging reduces net interest income volatility to 3.3%. Deposit non-interest income volatility is similar at 5.0%, and less than half that of *ex-deposit* non-interest income (11.7%), which includes investment banking and other non-traditional activities. These patterns are consistent across sub-periods with different interest rate and regulatory environments. Overall, the summary statistics demonstrate that deposit non-interest income is an important and stable source of bank income.

TABLE 2: AGGREGATE INCOME GROWTH VOLATILITY: 2000-2024

	Interest			Non Interest		Op. Exp.
	Income	Expense	Net Income	Deposit	Ex-Deposit	
Total Sample	5.2	20.9	3.3	5.0	11.7	4.3
2000-2010	4.8	10.2	3.3	3.8	16.7	4.6
2011-2024	5.5	26.6	3.3	4.9	6.1	3.5

Notes: Income is measured in real terms, using the PCE price deflator. Volatility is measured from the aggregate real quarterly growth rate of each series, measured in percentage points. In addition, the time series are filtered using a moving average method to remove seasonality. Operating expenses are measured based upon total non-interest expense. Sample from 2000-2025Q1. Source: Call Report.

Account Pricing and Bank Characteristics. For a typical deposit account, banks earn both non-interest income from fees as well as interest income from deposit spreads. Splitting the sample into *high* and *low* deposit NII banks, Table 3 documents that high deposit NII banks annually earn \$75 in non-interest income per account relative to \$39 for low deposit NII banks. For interest income, high deposit NII banks earn a 93 basis point deposit spread relative to an 85 basis point spread for low deposit NII banks; thus, high deposit NII banks generate more income from fees *and* rates.

To understand the total cost to depositors, we construct a quarterly bank-level measure p^d for the *effective deposit account price* defined as:

$$p^d = \frac{\text{deposit spread} \times \text{deposits} + \text{deposit NII}}{\text{number of accounts}} \quad (1)$$

$$= \text{deposit spread} \times \text{account size} + \text{account fees},$$

which captures foregone interest income plus fees for a deposit account. The average high deposit NII bank has an annual account price of \$199 relative to \$170 for a low deposit NII bank; thus, high deposit NII banks generate more income from deposit accounts. We also find that high deposit NII banks operate with lower deposit betas, implying that deposit rates are less sensitive to changes in the short rate, which we examine more rigorously in Section 3.2.

Consistent with d’Avernas et al. (2023), we find that bank size is relevant for account pricing. Large banks (assets $\geq$ \$250b) set higher fees and deposit spreads compared to medium banks (assets between \$10b and \$250b) and small banks (assets $<$ \$10b), resulting in higher effective account prices. In terms of the balance sheet, high NII banks fund their assets with more deposits and higher leverage, captured by higher deposit-to-asset ratios and lower tangible common equity (TCE) ratios, respectively. Thus, high NII banks attract more deposits despite charging higher account prices, which introduces *greater* bank risk as evidenced by lower z-scores.⁷ In terms of profitability, returns are relatively similar across both bank types, as measured by return on equity (ROE) and return on assets (ROA). This suggests that even though high NII banks set higher account prices, differences in operating expenses can be an important factor. For example, we observe that high NII banks spend more on data processing and marketing, which likely offsets some of the income from higher prices.

⁷Z-scores are computed by summing a bank’s capital ratio and return on assets, and dividing by the volatility of return on assets, yielding a normalized measure of solvency risk. For example, see Lapetit and Strobel (2015) or Bertray, Demirguc-Kunt, and Huizinga (2013).

TABLE 3: EMPIRICAL MOMENTS: 2000 Q1-2024 Q1

Moment	Units	Non-Int Share		Bank Type		
		High	Low	Large Bank	Medium Bank	Small Bank
Deposit NII-per-Account	\$	75	39	84	78	57
Deposit Spread	%	0.93	0.85	1.06	0.97	0.89
Deposit Account Price	\$	199	170	351	337	181
Deposit Beta	—	0.296	0.319	0.183	0.376	0.306
Real Assets	\$b	2.7	1.7	884	44	0.43
TCE Ratio	%	10.15	12.87	6.92	8.94	11.53
RW Capital Ratio	%	17.6	22.9	14.1	16.0	20.3
Z-score	—	22.1	23.0	15.0	16.0	22.6
Deposit-to-Asset	%	84.2	72.1	72.0	67.5	78.4
Dividend Yield	%	0.69	0.58	0.70	0.67	0.49
ROE	%	10.0	9.6	10.1	9.9	9.2
ROA	%	0.98	1.00	0.96	1.06	0.96
Expense on Data Processing	%	5.2	4.7	3.8	3.1	5.0
Expense on Marketing	%	1.5	1.4	2.2	2.4	1.4
Total NII Share	%	29.5	25.4	33.8	27.5	14.1
Deposit NII Share	%	6.7	1.3	5.2	4.0	4.5

Notes: Unless otherwise specified, all objects are annualized and computed as asset-weighted averages. *Deposit NII Share* measures non-interest income using only the deposit service charge line item. The *TCE Ratio* represents the tangible common equity ratio, which accounts for unrealized losses. The *Deposit Beta* is a bank-level measure of the change in deposit rates relative to the change in the Federal Funds rate. Both expense measures calculate the share of that item relative to the total operating cost of the bank. Bank sizes correspond to average, real asset thresholds where *Small Banks* are less than \$10 billion, *Medium Banks* are between \$10 billion and \$250 billion, and *Large Banks* are greater than \$250 billion. Source: Call Report.

Account-level Pricing. We find evidence that deposit NII is largely invariant to account size; that is, deposit NII behaves like a lump-sum account fee. Appendix Figure A.3 plots deposit NII per account against average account size in a bank-quarter binned scatterplot. The relatively flat relationship suggests that non-interest income is *not* proportional to deposits in an account.

Deposit Costs. We next estimate empirical cost functions and construct deposit markups to determine whether high non-interest income banks exhibit pricing power. We estimate a translog cost function similar to Berger et al. (2017), Berlin and Mester (2015), and Corbae and D’Erasmus (2021). Under this specification, the cost function can be expressed as $Cost_{it} = F(\mathbf{q}_{it}, \mathbf{w}_{it}, \mathbf{x}_{it}, \epsilon_{it}; \boldsymbol{\beta})$, where log cost for bank i at time t is linear in a vector of log-transformed bank outputs $\mathbf{q}_{it}$, input prices $\mathbf{w}_{it}$, interactions $\mathbf{x}_{it}$, and an error term ϵ_{it} . We specify total lending (q^ℓ) and insured deposit accounts (q^d) as outputs, and three input prices: labor wage cost (w^1), interest expense (w^2), and physical capital (w^3).⁸ We also include bank and time fixed effects.⁹

⁸We focus on insured deposit accounts, as data on the number of accounts are only available for this category. We also believe this category accurately reflects the type of consumer accounts that are related to deposit NII.

⁹See Section B for details of the regression specification and estimation.

We define cost as net non-interest expense but also consider gross non-interest expense for robustness. Importantly, because our analysis focuses on account-level pricing, we estimate the cost function with respect to the *number of deposit accounts* but also consider specifications with the *quantity of deposits*. Once estimated, we derive marginal cost $\widehat{mc}_{it}^{deposit}$ and calculate the markup $\frac{p_{it}^d - \widehat{mc}_{it}^{deposit}}{\widehat{mc}_{it}^{deposit}}$ where p_{it}^d is the effective deposit account price defined in Equation (1).

Table 4 shows that, on average, banks incur a marginal cost of \$118 per account. Marginal cost for high NII banks is \$93 compared to \$141 for low NII banks. Given their higher account prices and lower marginal costs, high NII banks earn a markup of 107% versus 28% for low NII banks. We perform a similar estimation at the deposit level and find that costs are more similar across bank types, with high NII banks paying 1.27¢ per deposit relative to 1.24¢ for low NII banks.¹⁰ Taken together, we interpret these findings as strong evidence that high non-interest income banks exert pricing power in the deposit market.¹¹

TABLE 4: ESTIMATED BANK MARGINAL COST AND MARKUPS

Object	All Banks	High NII Banks	Low NII Banks
Account Marginal Cost	\$118/Acct	\$93/Acct	\$141/Acct
Account Markup	67%	107%	28%
Deposit Marginal Cost	1.26¢/Deposit	1.27¢/Deposit	1.24¢/Deposit

Notes: Marginal costs are reported as bank-quarter averages of the marginal cost formula in Equation (18) of Appendix Section B and reported in \$ terms at the account level. High NII banks correspond to banks with above median deposit NII as in Table 3. Markups are computed as the ratio of $p^d - \widehat{mc}^{deposit}$ to marginal cost $\widehat{mc}^{deposit}$ where the deposit account price corresponds to Equation (1).

If high non-interest income banks earn higher markups, why are total bank profits similar across bank types? One possibility is that our cost estimation captures only *contemporaneous* costs and not past investments that may affect future profits. For example, a bank may invest in its technology *today* to increase consumer demand and markups *tomorrow*.¹² We estimate a local projection model and find that a one standard deviation increase in data spending is associated with a 9.3% cumulative increase in deposit account prices over five years (Figure B.1 in Appendix

¹⁰Wang et al. (2022) report a marginal cost of 0.9¢/Deposit. Egan et al. (2017) report a marginal cost of 5¢ to 5.5¢. Hanson et al. (2015) and Drechsler et al. (2025) use hedonic cost regressions and recover estimates of 1.55¢ and 1.3¢ for insured deposits, respectively.

¹¹We also estimate a hedonic cost regression in Appendix Table B.2. We find that high NII banks exhibit lower marginal cost for uninsured deposits and equal marginal cost for insured deposits, which implies higher markups.

¹²We find no evidence of serial correlation in our cost estimation, suggesting that past spending improves pricing only through the demand channel and not through cost efficiency.

Section B). Thus, past investments are a potential mechanism for pricing power among high NII banks, although we take the source of this pricing advantage as exogenously given when modeling banks in Sections 4 and 5.

3 Empirical Evidence on Monetary Policy Transmission

This section uses branch-level and bank-level data to present empirical evidence on the relationship between monetary policy, deposit non-interest income, fees, and deposit rates over time. We provide evidence that banks with higher deposit non-interest income exhibit deposit pricing power via a lower pass-through of monetary policy to deposit rates. We do not find that deposit non-interest income meaningfully responds to monetary policy, suggesting that deposit fee pricing is rather sticky and invariant to monetary policy.¹³

3.1 Empirical Strategy

We use state-dependent local projections to assess the pass-through of monetary policy to deposit rates across banks with different non-interest income shares. At each horizon h , we estimate the following regression equation for bank i 's branch c at time t :

$$r_{t+h,i,c} - r_{t-1,i,c} = \alpha_{i,c}^h + \beta^h s_t + \gamma^h s_t \times NII_{t,i} + \theta^h X_{t,i} + \eta^h Z_t + \epsilon_{t+h,i,c}, \quad (2)$$

where $r_{t+h,i,c} - r_{t-1,i,c}$ represents the cumulative rate change, $\alpha_{i,c}^h$ reflects a bank-branch fixed effect, s_t stands for a standard monetary policy shocks taken from Jarociński (2024),¹⁴ $NII_{t,i}$ reflects the five-year average deposit NII share,¹⁵ and $Z_{t,i}$ is a vector of macroeconomic and financial controls. The vector of macroeconomic and financial controls contains the unemployment rate, industrial production growth rate, CPI inflation, VIX, excess bond premium, and a dummy for the zero lower bound period. As a robustness check, we control for size (log assets), capitalization (equity ratio), maturity gap based on English et al. (2018), wholesale funding ratio, and other non-interest

¹³On average, bank-level fee price drops by up to 4% in response to a monetary shock scaled to a 100 bp increase in the policy rate (Figure A.6). Monthly fee charges for saving and money market accounts show little variation over time (Figure A.2).

¹⁴The standard monetary policy shock captures unexpected movements in the short-end of the yield curve and isolates any information effect. The monetary policy shock is scaled to increase the federal funds rate by one percentage point after 12 months.

¹⁵Using the five-year lagged average instead of the contemporaneous share addresses the potential endogeneity concern that banks actively adjust their deposit NII share to monetary policy.

income in those specifications with bank-level controls and interact these with the monetary policy shocks (Appendix Figures A.4 and A.5).¹⁶

The pass-through of monetary policy to deposit rates is defined as the derivative of the change in the deposit rate to the monetary policy shocks and corresponds to the sum of $\beta^h + \gamma^h X_{t,i}$. For the visualization of the results, we focus on two states of $X_{t,i}$: low and high shares defined as the 10th and 90th percentile of the deposit non-interest income share distribution.

3.2 Empirical Results

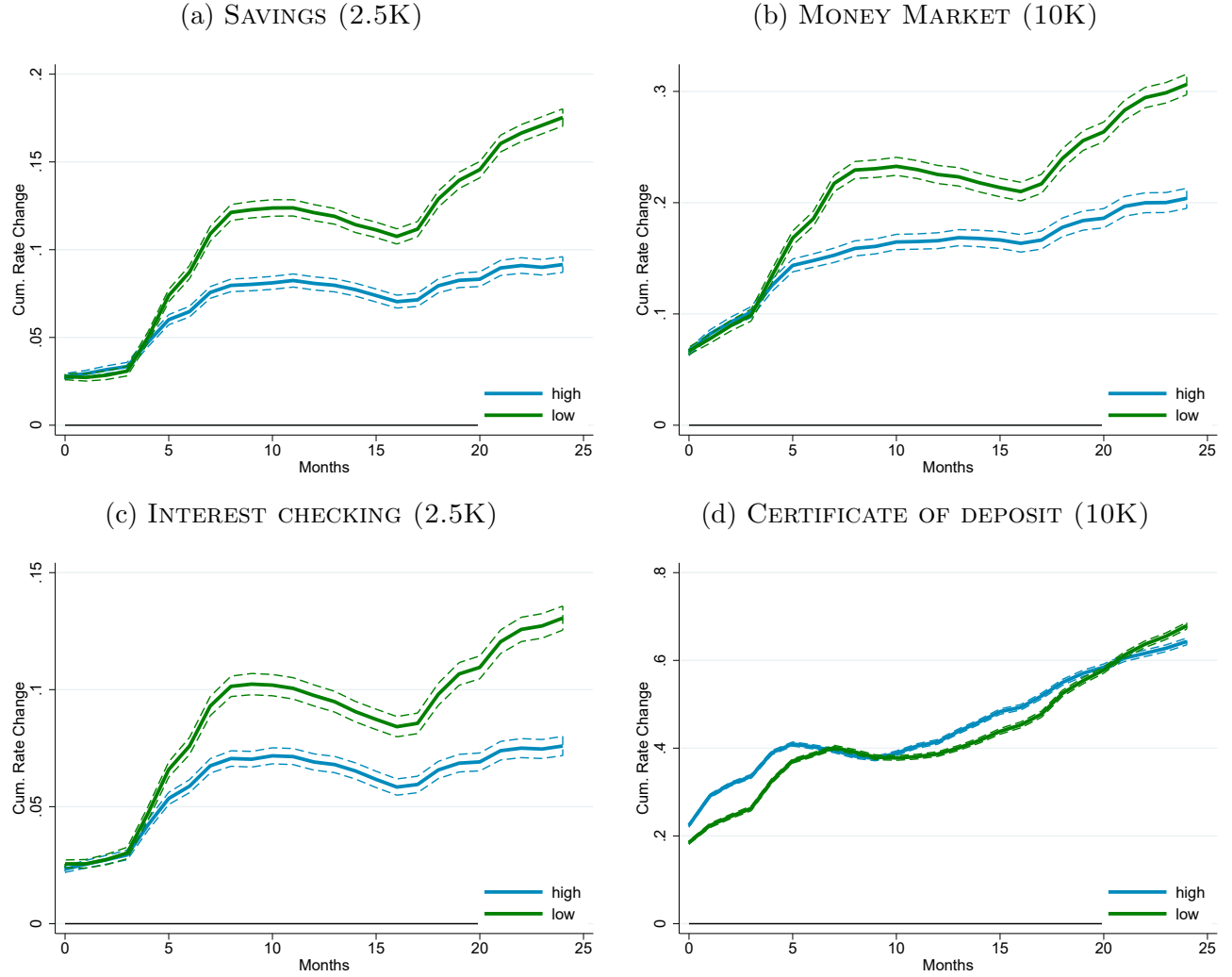
Figure 2 shows that the estimated pass-through of monetary policy shocks to deposit rates is related to bank deposit non-interest income shares. The figure plots pass-through for four different types of accounts: savings accounts, money market accounts, interest checking accounts, and certificates of deposit. The key finding is that deposit rate pass-through is lower for banks with higher deposit non-interest income shares. The results are most pronounced for savings and interest checking accounts, which are more clearly linked to depositor liquidity services. Conversely, certificates of deposit (CDs), which offer a very limited set of services to depositors, show little difference in pass-through. Similar patterns emerge across a wider range of deposit accounts (Appendix Figures A.7 and A.8). Consistent with these findings, bank-level estimates using imputed deposit rates show that savings and money market rates increase less in response to monetary policy shocks at high-NII banks (Figure A.9).

Even though high deposit non-interest income banks exhibit lower pass-through, they experience similar or less deposit outflows when compared to low deposit non-interest income banks (Figure A.10). Thus, they are able to maintain similar levels of funding but at a cheaper rate. Consistently, their net interest margins respond less to monetary policy (Figure A.10). To sum up, banks with greater dependence on fee income (as proxied by deposit non-interest income shares) exhibit pricing power as measured by low rate pass-through.

Bank-level evidence on rates. To address concerns about aggregation, compositional effects, or the influence of new deposits (since RateWatch reports offered rates), we use quarterly bank-level Call Report data as an additional source. These data include all deposit accounts and are weighted by actual balances. Using reported interest expenses and balances, we impute bank-level rates for

¹⁶We use the deviation from the period average to account for secular trends in the interaction variables.

FIGURE 2: LOCAL PROJECTIONS OF DEPOSIT RATES TO MONETARY POLICY SHOCKS



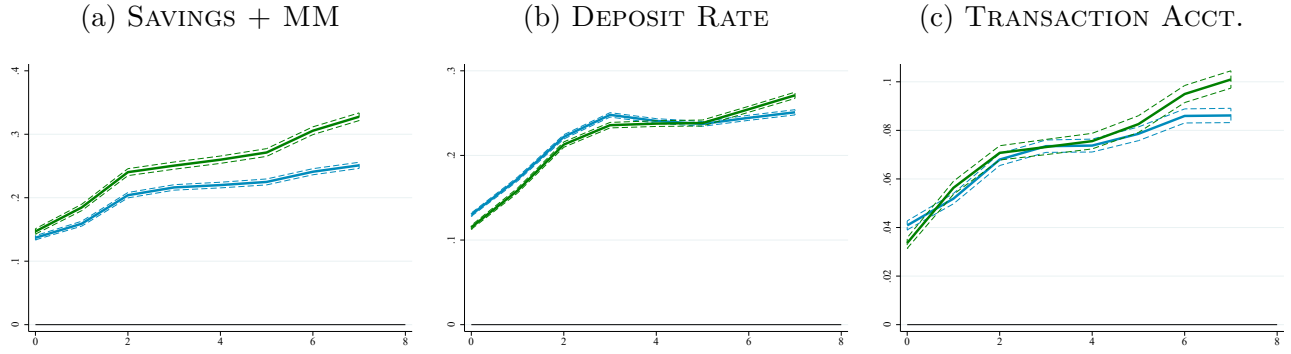
Notes: Impulse responses of deposit rates (savings accounts, money market accounts, interest checking accounts, and certificates of deposit) to a monetary policy shock at both high (blue) and low (green) shares of deposit NII to interest rate income. Horizon is in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

total deposits, transaction accounts, and savings accounts.

Similar to the branch-level evidence, Figure 3 shows that banks with a higher share of deposit NII exhibit lower pass-through for savings accounts (including money market accounts), higher pass-through for overall deposit rates (which include time deposits such as CDs), and no significant difference for transaction accounts.

FIGURE 3: BANK-LEVEL LOCAL PROJECTIONS BY DEPOSIT NII

Panel A: Without controls



Notes: Impulse responses of different bank-level deposit rates to a monetary policy shock at both high (blue) and low (green) shares of deposit NII. Horizon is in quarters. Standard errors are clustered by bank. 90% confidence intervals. Source: Call Report.

Other proxies. To provide additional support for the main empirical finding, we perform several robustness checks: (i) using account fees in place of non-interest income shares, (ii) using the log number of branches in place of non-interest income shares, and (iii) examining the pass-through behavior of internet banks, which offer a limited range of liquidity services. We find consistent evidence that banks with higher account fees (Figure A.11), banks with larger branch networks (Figure A.12 and Figure A.13), and brick-and-mortar banks (Appendix Figures A.14 - A.16) exhibit lower deposit rate pass-through of monetary policy shocks. Results for account fees confirm that banks more reliant on non-interest income exhibit lower rate pass-through.¹⁷ We interpret larger branch networks and brick-and-mortar banking as proxies for broader liquidity services provision, suggesting that service quality is an important determinant of pass-through and pricing power.¹⁸

¹⁷Account fees are measured as total service charges on deposit accounts relative to deposits, although our preferred measure of fees is the non-interest income share, as it is less subject to measurement error than price data.

¹⁸The sample of internet banks is too small to allow state-dependent local projections. Instead, we compare the pass-through of monetary policy to deposit rates across bank types.

Bank concentration vs. non-interest income Does non-interest income matter beyond the traditional deposit channel of monetary policy transmission—that is, after controlling for local market concentration? To address this question, we estimate regressions in the spirit of Drechsler et al. (2017), contrasting the roles of market concentration and non-interest income in the pass-through of monetary policy to various deposit rates. Specifically, we estimate the change in the deposit rate $\Delta r_{t,i,c}$ to monetary policy conditional on local concentration and share of deposit non-interest income at the branch and bank levels:

$$\Delta r_{t,i,c} = \alpha_c + \zeta_{c(i)} + \eta_t + \beta \Delta FF_t + \gamma_1 \Delta FF_t \times HHI_{t-1,c} + \gamma_2 \Delta FF_t \times NII_{t,i} + \varepsilon_{t,i,c}, \quad (3)$$

$$\Delta r_{t,i} = \zeta_i + \eta_t + \beta \Delta FF_t + \gamma_1 \Delta FF_t \times HHI_{t-1,i} + \gamma_2 \Delta FF_t \times NII_{t,i} + \varepsilon_{t,i}, \quad (4)$$

where $\Delta r_{t,i,c}$ denotes the change in the respective deposit rate, ΔFF_t is the change in the federal funds rate, $HHI_{t-1,i,c}$ is the concentration measure, the lagged county-level Herfindahl-Hirschman-Index (HHI),¹⁹ $NII_{t,i}$ is the five-year average of the deposit non-interest income share of bank i , and α_c and $\zeta_{c(i)}$ are county and branch fixed effects, respectively.

Table 5 and Appendix Table A.1 report the results across three deposit products at the bank and branch level, respectively. The interaction between the federal funds rate change and non-interest income share γ_2 is negative and statistically significant for saving rates, indicating lower deposit rate pass-through for banks with higher reliance on non-interest income. The effect is economically meaningful and consistent across all product types. A 1 p.p. increase in the deposit non-interest income share is associated with a reduction in pass-through by -0.1 to -0.5 on impact. Notably, the magnitude is independent of whether the regression is jointly estimated with concentration or solely with the non-interest income share.

Deposit betas. To address the potential concern that rate dynamics and spreads may be influenced by the overall level of interest rates, we follow Drechsler et al. (2017) and Drechsler et al. (2021) and estimate *deposit betas* that measure how deposit rates respond to changes in the federal funds rate. Using both branch-level and bank-level data, we estimate deposit rate betas based on time-series regressions. For each branch or bank i , we estimate the response of deposit rates to the federal funds rate:

¹⁹For the bank-level regressions, we calculate for each bank a deposit-weighted bank-HHI based on county-level HHIs.

TABLE 5: RATE PASS-THROUGH, CONCENTRATION AND NON-INTEREST INCOME

Panel A: Savings rates			
HHI $\times$ dFF	-0.143*** (-3.55)		-0.0983** (-3.16)
Deposit NII $\times$ dFF		-0.557** (-2.96)	-0.498* (-2.64)
N	530609	491710	491689
R^2	0.264	0.265	0.266
Panel B: Transaction rates			
HHI $\times$ dFF	0.0312 (1.30)		0.0332 (1.28)
Deposit NII $\times$ dFF		-0.0447 (-0.44)	-0.0652 (-0.57)
N	499156	465178	465161
R^2	0.127	0.134	0.134
Panel C: Deposit rates			
HHI $\times$ dFF	-1.615 (-1.13)		-2.284 (-1.12)
Deposit NII $\times$ dFF		10.88 (1.29)	12.29 (1.26)
N	579940	537542	537016
R^2	0.0395	0.0424	0.0425

Notes: This table reports estimated pass-through of policy rate changes to deposit rate changes conditional on bank-level deposit market concentration (HHI) and banks' deposit-related non-interest income (NII) share, both estimated separately and jointly in one regression. Panel A reports results for savings rates, Panel B for transaction rates, and Panel C for composite deposit rates. All specifications include lagged dependent variables, branch, bank, and time fixed effects, as well as macroeconomic controls. Standard errors are clustered at the bank and quarter level. *, **, and *** indicate significance at the 5%, 1%, and 0.1% levels.

$$\Delta r_{it}^d = \alpha_i^I + \beta_i^I \Delta F F_t + \epsilon_{it}, \quad (5)$$

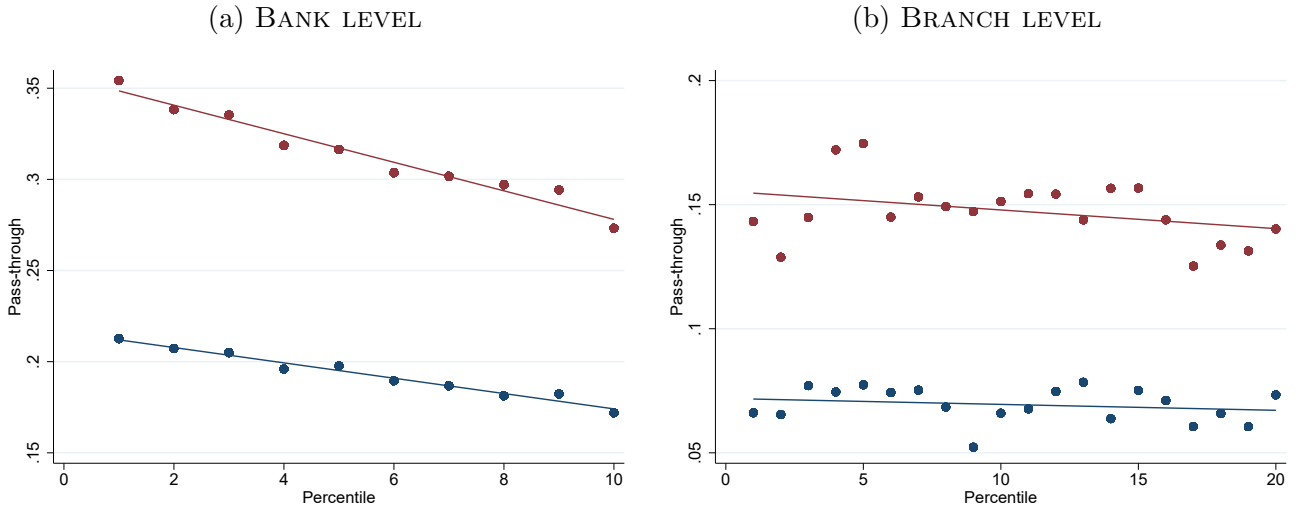
as well as the response to changes in the federal funds rate over the past year:

$$\Delta r_{it}^d = \alpha_i^C + \sum_{i=0}^{12/4} \beta_i^C \Delta F F_{t-i} + \epsilon_{it}, \quad (6)$$

where Δr_t^d reflects the change in the deposit rate from date t to $t + 1$, and $\Delta F F_t$ is the change in the federal funds target rate from t to $t + 1$. We estimate both specifications at the bank and branch levels, excluding banks with too few observations or insignificant estimates. In the branch-level analysis, we average estimated betas across branches within each bank, sort banks into 20 and 10 bins based on their deposit NII share, and compute average betas for each bin.

Figure 4 plots the relationship between deposit betas and the share of deposit NII for saving accounts. Panel a shows results at the bank level, and Panel b shows results at the branch level. In both panels, blue dots and lines correspond to contemporaneous deposit betas, while red dots and lines correspond to cumulative betas over the past year. Deposit betas tend to decline with the share of deposit NII, indicating that banks and branches with a higher reliance on deposit NII are associated with a lower pass-through, i.e., adjust deposit rates less in response to changes in the federal funds rate. The decline is more pronounced at the bank level than at the branch level. Figure A.17 provides robustness for different account types.

FIGURE 4: BIN SCATTER DEPOSIT BETA BY DEPOSIT NII SHARE (SAVINGS ACCOUNT)



Notes: This figure shows binned scatter plots of deposit rate betas estimated at the bank level (left panel) and branch level (right panel) by deposit non-interest income share. The blue dots and line report the contemporaneous effects, and the red dots and line report cumulative effects over 12 months.

Other bank-level outcomes. Appendix Figure A.10 shows the responses of bank-level deposit flows, lending, assets, net interest margins, and profitability (ROA, ROE) to monetary policy shocks for banks with high and low deposit NII. Profitability rises in response to a monetary tightening, particularly for high deposit NII banks, reflecting their larger increase in deposit spreads. Deposits, assets, and lending contract modestly, with few economically significant differences across bank types. For all banks, wider deposit spreads induce deposit outflows and reduce resources available for lending. Higher lending rates (Appendix Figure A.18) further contribute to the contraction in lending. Relative to lending, we observe a larger contraction in mortgage-backed securities (MBS) and US government agency obligations in response to monetary tightening (Appendix Figure A.19).²⁰

4 Illustrative Model

To help rationalize the main empirical findings, we develop a simple model of two-part bank pricing in which the deposit rate and account fee govern the intensive and extensive margins of demand, respectively.²¹ Banks with greater deposit market power are able to set higher prices, exhibit lower rate pass-through, and operate with larger non-interest income shares.

Consumers have preferences over consumption and deposits $u(c, d) = c + v(d, \theta, r)$, where $\theta \sim U[0, \Theta]$ is a heterogeneous taste parameter and r is an available market return. We assume that depositor preferences $v(d, \theta, r)$ are concave in deposits and increasing in θ and r .²² For tractability, we assume $v(d, \theta, r) = a(r)\theta d - \frac{d^2}{2}$, where $a(r)$ is an increasing function that can be interpreted as a depositor's outside option becoming less attractive as r rises, similar to the role of cash in Drechsler et al. (2017).

With an endowment ω , depositors allocate funds between a market return r and deposits with return r^d . Opening an account requires a lump-sum account fee f . Thus, the depositor budget constraint can be written as:

$$c = (1 + r)\omega - \underbrace{\left(sd + f \mathbb{I}[d > 0] \right)}_{p^d}, \quad (7)$$

²⁰For example, see Drechsler et al. (2024).

²¹The model is one of nonlinear pricing with two-part tariffs and heterogeneous demand, similar to Tirole (1988).

²²Deposits in the utility function capture the liquidity services or benefit that banks provide consumers.

where p^d is the effective deposit account price as in Equation (1) and $s = r - r^d$ is the deposit spread. Conditional on opening an account, type- θ deposit demand is:

$$d^\theta(r, s) = a(r)\theta - s, \quad (8)$$

which is linear in the deposit spread and shifts with the level of interest rates r .

Given the distribution of depositors, a monopolistic bank chooses a deposit spread s and fee f to maximize profits $\Pi = (s - mc)D(s, f, r) + fN(s, f, r)$, where $D(s, f, r)$ is total deposit demand, $N(s, f, r)$ is the mass or number of accounts, and mc is the marginal operating cost. We interpret the exogenous distribution parameter Θ as a measure of bank market power—higher Θ represents a decrease in deposit demand elasticity. Proposition 1 summarizes equilibrium pricing and its relation to market power.²³

Proposition 1. *Given $a(r)\Theta > 2mc$, there exists an equilibrium threshold $\bar{\theta}^*$ such that depositors of type $\theta \geq \bar{\theta}^*$ choose to open an account. The bank sets an equilibrium deposit spread and fee*

$$\begin{aligned} s^* &= \frac{4}{5}mc + \frac{a(r)}{5}\Theta \\ f^* &= \frac{2}{25}(a(r)\Theta - mc)^2 \end{aligned}$$

where $\bar{\theta}^* = \frac{2}{5}\frac{mc}{a(r)} + \frac{3}{5}\Theta$.

Proposition 1 demonstrates that bank pricing is determined by two factors—marginal cost and market power Θ . A bank facing deposit demand with support $[0, \Theta']$ and $\Theta' > \Theta$ is pricing against a less elastic demand curve, and this confers a pricing advantage for the bank. Consistent with the broader class of two-part pricing models, banks use account fees to extract consumer surplus, affecting the decision to open an account, while the deposit spread controls deposit intensity for the inframarginal depositor. Proposition 2 summarizes the relationship between market power, bank pricing, and non-interest income shares.

Proposition 2. *Banks with increasing deposit market power, as characterized by $\Theta' > \Theta$, exhibit*

(i) *Higher effective deposit account prices (i.e. $\frac{\partial p^{d*}}{\partial \Theta} > 0$) for all $\theta \in [\bar{\theta}^*, \Theta]$*

(ii) *Higher deposit spreads (i.e. $\frac{\partial s^*}{\partial \Theta} > 0$) and higher account fees (i.e. $\frac{\partial f^*}{\partial \Theta} > 0$)*

²³Refer to Section C for explicit model derivations.

(iii) *Higher non-interest income shares (i.e. $\frac{\partial NII^*}{\partial \Theta} > 0$)*

(iv) *Lower rate pass through (i.e. $\frac{\partial^2 s^*}{\partial \Theta \partial r} > 0$)*

Proposition 2 rationalizes the main empirical finding of the paper—namely that banks with higher non-interest income shares set higher deposit spreads, higher account fees, and exhibit lower rate pass-through. Importantly, the proposition shows that variation in deposit market power is the underlying source of variation in bank pricing and non-interest income. Put differently, high non-interest income is a symptom of deposit market power in this framework. The intuition for why non-interest income shares increase with market power is that banks can more efficiently extract surplus through fees, which are non-distortionary; thus, non-interest income rises more quickly than interest income as market power increases.²⁴

5 Quantitative Model

We next calibrate a quantitative model of the US banking sector, drawing on the insights from the illustrative model to inform reduced-form assumptions regarding deposit demand. We then use the quantitative model for counterfactual analysis, examining the implications of non-interest income for credit supply, bank risk, and deposit pricing.

5.1 Bank Problem

There exist I banks that monopolistically compete for deposits. Banks generate interest income from a standard loan-deposit business model but also generate non-interest income from deposit account fees.

Deposit Demand. Total bank deposits D_i are given by the product of deposit intensity d_i , which is the number of deposits per account, and the number of accounts N_i such that $D_i = N_i d_i$. Each period, banks borrow deposits D_i at a rate r_i^d while also earning non-interest income $f_i N_i$ from account fees f_i . To maintain tractability, we assume that the intensive margin of demand $d_i(s_i, r)$ depends on the deposit spread $s_i = r - r_i^d$ and the monetary policy rate r , whereas the extensive margin of demand $N_i(f_i)$ depends only on the account fee f_i . This specification is motivated by the

²⁴For a more mechanical understanding, observe that equilibrium fees scale quadratically with deposit market power in Proposition 1.

illustrative model, where deposit demand responds only to spreads and the level of interest rates for inframarginal depositors, while account fees serve as the primary screening device governing the extensive margin.²⁵

Technology. Banks originate one-period loans ℓ_i which generate an exogenous, risky return r^ℓ such that loan returns are:

$$r^\ell = r + \Delta^\ell \quad (9)$$

where r is the monetary policy rate, determined by a Taylor-type monetary policy rule, and Δ^ℓ is a constant spread meant to capture a positive term premium. The monetary policy rate is determined by $1 + r' = (1 + \bar{r})(\frac{z'}{z})^{\phi_z} \epsilon'_r$, where $\bar{r}$ is a risk neutral rate, z is an aggregate productivity process which reflects business cycle fluctuations, and ϵ_r is an iid, mean one monetary policy shock process with standard deviation σ_r . We assume z follows an AR(1) process such that $z' = \rho_z z + \epsilon'_z$ where ϵ_z is an iid, mean zero random variable with standard error σ_z .

Constraints. Banks enter each period with some endogenous level of network n_i and must satisfy a budget constraint:

$$\pi_i + \ell_i + C_i(\pi_i, \ell_i, N_i) = n_i + D_i + f_i N_i, \quad (10)$$

where the right-hand side represents bank funding, consisting of network n_i , total deposits D_i , and non-interest income $f_i N_i$. Bank funding is used to distribute dividends π_i , originate new loans ℓ_i , and pay operational costs $C_i(\pi_i, \ell_i, N_i)$ related to equity issuance, loan origination, and the servicing of deposit accounts.

The dynamics of the bank problem are captured through the law of motion for next-period

²⁵Generally, both spreads and fees affect the account-opening decision in the illustrative model. Proposition 3 in Appendix Section C establishes sufficient conditions under which depositor participation is more sensitive to account fees than to spreads. These conditions can be expressed in terms of equilibrium non-interest income shares remaining below a threshold that is consistently satisfied in the data. Thus, we view the reduced form assumption that account demand depends only on fees as a tractable approximation.

network n'_i :

$$\begin{aligned} n'_i &= (1 + r^{\ell'})\ell_i - (1 + r_i^d)D_i \\ &= (1 + r' + \Delta^\ell)\ell_i - (1 + r_i^d)d_i(s_i, r)N_i(f_i), \end{aligned} \quad (11)$$

such that the main source of stochastic variation in bank network is driven by the monetary policy rate due to both business cycle fluctuations (i.e. the productivity process z) and pure monetary policy shocks. These stochastic processes have a direct effect on both loan returns and deposit demand.

Optimization. The state of bank i is determined by its network n_i , the aggregate state z , and the monetary policy rate r . The bank's objective is to maximize the expected, discounted dividend stream to equity owners such that the bank i value function can be expressed as:

$$\begin{aligned} v_i(n_i, z, r) &= \max_{\pi_i, r_i^d, f_i, \ell_i} \pi_i + \beta E[v_i(n'_i, z', r')] \\ \text{s.t.} \quad &\text{Eqns (9) -- (11)} \end{aligned} \quad (12)$$

where the bank chooses dividends π_i , the deposit rate r_i^d , account fees f_i , and total lending ℓ_i . Refer to Section D for an explicit characterization of bank equilibrium conditions.

5.2 Calibration

Functional Forms. Motivated by the micro-founded approach in Section 4, we model the intensive margin of deposit demand as:

$$d_i(s_i, r) = \bar{q}_i(r) - \alpha_i^d s_i, \quad (13)$$

such that demand is linear in spreads and changes in the monetary policy rate shift the demand curve. The demand function is parameterized with spread sensitivity $\alpha_i^d > 0$ and $\bar{q}_i(r) = \frac{\bar{q}_i}{1 + e^{-\xi_i r}}$ where $\xi_i > 0$. For the extensive margin, we choose account demand $N_i(f_i) = e^{-\alpha_i^n f_i}$ with fee sensitivity $\alpha_i^n > 0$ such that demand is bounded on the unit interval.

Operating costs of the bank are expressed as:

$$C_i(\pi_i, \ell_i, q_i^n) = \frac{\phi_i^\pi}{2}(\pi_i - \bar{\pi}_i)^2 + \frac{\phi_i^\ell}{2}\ell_i^2 + \phi_i^n N_i, \quad (14)$$

which includes a dividend adjustment cost, a convex loan origination cost, and a linear cost in deposit accounts. Dividend adjustment costs are an important financial friction, affecting the rate at which banks substitute between debt and equity over time and, thus, are an important determinant of bank leverage and equity issuance. The quadratic loan origination cost indicates decreasing returns to scale loan technology and introduces concavity to the bank problem, affecting decisions regarding balance sheet risk and expected returns. Consistent with empirical findings in Section 2, we assume that banks incur deposit costs (e.g., expenses on marketing or data).

Equilibrium Deposit Pricing. The model departs from the standard loan-deposit framework by including account fees and separate account demand. Optimal pricing on the intensive margin remains relatively simple such that the deposit rate can be expressed as:

$$1 + r_i^d = \frac{1}{E[m'_i]} - \frac{d_i}{\partial d_i / \partial r_i^d} \quad (15)$$

where m'_i is the bank's stochastic discount factor and the term $\frac{d_i}{\partial d_i / \partial r_i^d}$ represents a markdown, allowing the bank to charge a positive deposit spread.²⁶ Somewhat differently, optimal fees can be expressed as:

$$f_i = \underbrace{\phi_i^n + \frac{N_i}{\partial N_i / \partial f_i}}_{\text{mc + markup}} - \underbrace{\frac{E[m'_i] d_i^2}{\alpha_i^d}}_{\text{intensive margin value}} \quad (16)$$

which contains a cross-subsidizing feature with respect to the intensive margin. Specifically, while banks set a fee markup above the marginal cost of servicing their accounts, they also set lower fees if the value of the intensive margin is high.

External Calibration. Model parameters can be summarized by the bank i -specific set:

$$\Theta_i = \left\{ \underbrace{\bar{r}, \phi_z, \sigma_r}_{\text{monetary policy}}, \underbrace{\rho_z, \sigma_z, \Delta^\ell}_{\text{loan returns}}, \underbrace{\phi_i^\pi, \bar{\pi}_i, \phi_i^\ell, \phi_i^n}_{\text{bank costs}}, \underbrace{\alpha_i^d, \bar{q}_i, \xi_i, \alpha_i^n}_{\text{demand}}, \underbrace{\beta}_{\text{Discount}} \right\} \quad (16)$$

where most cost and demand parameters are unique to a bank's specific type $i \in I$. Specifically, there are 8 bank-specific parameters $\{\phi_i^\pi, \bar{\pi}_i, \phi_i^\ell, \phi_i^n, \alpha_i^d, \bar{q}_i, \xi_i, \alpha_i^n\}$. We set $I = 2$, so that there exist two banks, and partition banks in the data according to their deposit non-interest income shares,

²⁶While N_i does not appear explicitly, it still affects the bank's stochastic discount factor m'_i .

referring to them as either the high deposit non-interest income (*high NII*) bank or low deposit non-interest income (*low NII*) bank. Table 3 provides empirical moments for each bank type.

Model parameters can be further partitioned into those which will be externally calibrated and those which will be internally calibrated; i.e., values determined by matching model moments with empirical counterparts. Parameters related to monetary policy $\{\bar{r}, \phi_z, \sigma_r\}$ are externally calibrated based upon common values within the literature. Further, we externally estimate the values of $\{\rho_z, \sigma_z\}$ by filtering real GDP and then setting the loan return spread Δ^ℓ to achieve an average spread observed in the data. We set the neutral rate of interest in the monetary policy rule to the inverse of the discount factor such that $\bar{r} = \beta^{-1} - 1$. External parameters are listed in Table 6.

Internal Calibration. The remaining structural parameters govern bank costs and deposit demand. We estimate these parameters by matching model moments to empirical targets that are informative about each parameter of interest. We employ the same set of target moments for each bank type $i \in \{high\ NII, low\ NII\}$ in the data.

For the intensive margin of deposit demand, spread sensitivity α_i^d has a first-order effect on how banks set their deposit spread $s_i = r - r_i^d$. We therefore calibrate α_i^d to match each bank's average deposit spread. While this is sufficient to capture the unconditional spread in the data, we are also interested in the dynamics of deposit pricing. The parameter ξ_i governs how deposit demand shifts with the level of the monetary policy rate and; thus, it affects the pass-through of policy rate changes to deposit rates. We therefore set ξ_i^d to target the average deposit beta. In addition, the demand intercept $\bar{q}_i$ determines the overall level of deposit demand, and we use it to match the average deposit-to-asset ratio on bank balance sheets.

We jointly calibrate fee sensitivity α_i^f and marginal cost ϕ_i^n parameters using two targets: the bank non-interest income share and the estimated deposit account markup $\hat{\mu}_i^n$ from Table 4. Specifically, we target bank non-interest income shares using fee sensitivity α_i^f . Because variation in non-interest income can reflect both variation in cost and demand, we exploit the steady-state version of the fee pricing condition in Equation (16) to pin down the cost parameter ϕ_i^n .²⁷ In the data, we include only non-interest income explicitly related to deposit accounts, taking this as a conservative lower bound for the income that banks generate from deposit relationships.

²⁷Specifically, for every candidate value of α_i^f , we set ϕ_i^n to satisfy the steady-state pricing condition and achieve the desired markup. The steady-state version of Equation (16) is $f_i = \phi_i^n - \frac{1}{\alpha_i^n}$, and given the estimated account markup $\hat{\mu}_i^n$, we pin down the marginal cost parameter via $\phi_i^n - \frac{1}{\alpha_i^n} = (1 + \hat{\mu}_i^n)\phi_i^n$.

TABLE 6: EXTERNAL CALIBRATION PARAMETERS

Parameter	Label	Value	Source/Target
β	Discount Factor	0.995	2% annual rate
ρ_z	Agg Shock Persistence	0.89	Fernald (2014)
σ_z	Agg Shock Volatility	0.0138	Real GDP Growth (1980s-Present)
ϕ_z	MP Exponential Term	0.9	Taylor (1993)
σ^r	MP shock term	40 bps	Coibion (2012), Christiano et al. (2005)
Δ^ℓ	Loan Spread	0.016	Corporate Loan Spread BofA

Notes: Model parameters are set for a model in quarterly frequency.

The dividend adjustment parameter ϕ_i^π affects the rate at which banks substitute between debt and equity funding over time. For example, when $\phi_i^\pi = 0$, the bank sets a constant deposit rate and all variation in bank networth and deposit demand is absorbed by the bank issuing dividends or equity. We set this parameter to target *dividend yield* defined as the ratio of dividends to bank assets. The parameter $\bar{\pi}_i$ is the long-run average dividend for each bank and, thus, is solved for internally within the calibration procedure. For loan cost ϕ_i^ℓ , we target the average marginal cost of lending for each bank type using moments from our empirical cost estimation in Section 2 and detailed in Appendix Section B.

Table 7 lists the internal model calibration and the fit of the model moments. Overall, the model achieves a strong fit with the data, particularly with respect to deposit pricing and non-interest income. The calibration also achieves intuitive cross-bank differences in parameter estimates. For example, high non-interest income banks have smaller spread and fee sensitivity parameters compared to low non-interest income banks, reflecting less elastic demand and greater market power.

Baseline Results and Validation. Appendix Table A.2 reports a set of untargeted model moments to assess the model’s fit. In terms of profitability, as measured by return on assets (ROA), the model provides reasonable approximations for both bank types. Fit is slightly weaker for the low NII bank because the calibrated model implies higher leverage, resulting in higher profitability relative to the empirical moment. We assess bank risk via z-scores. The model captures the qualitative pattern that banks with high non-interest income ratios are riskier. This result is primarily driven by their decision to hold more leverage (i.e., lower capital ratios), as reflected in

the deposit-to-asset ratios reported in Table 7. Because the level of fees charged by a bank is not directly interpretable within the model, we compare the gap between the fees set by high and low NII banks, which we refer to as the *Fee Difference*. In the model, fees charged by the high NII bank are 138% larger than those charged by the low NII counterpart, which is reasonably close to the empirical moment of 92.3%.

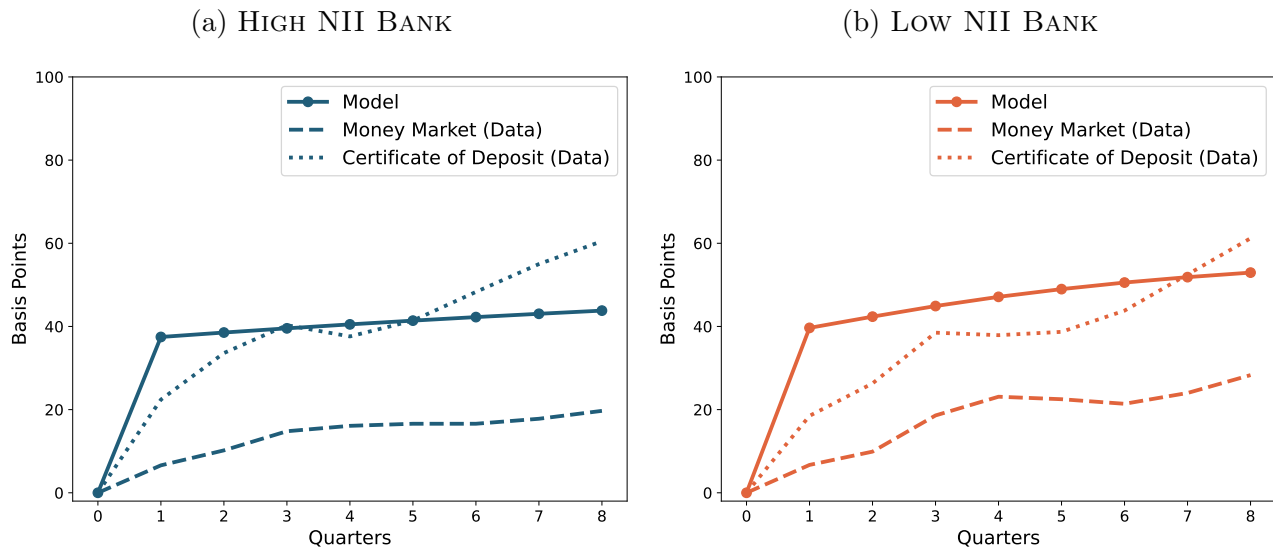
TABLE 7: INTERNAL CALIBRATION PARAMETERS

Parameter	Label	Value	Target	Data	Model
α_{high}^d	Spread Sensitivity	0.62	Deposit Spread	93.0	103.8
$\bar{q}_{high}$	Demand Intercept	1.99	Deposit-Asset Ratio	84	84
ξ_{high}	Deposit Shifter	0.81	Deposit Beta	0.30	0.35
ϕ_{high}^π	Dividend Adjustment	2.79	Dividend Yield	0.69	0.32
ϕ_{high}^ℓ	Loan Cost	0.02	Loan Marginal Cost	0.45	0.85
α_{high}^n	Account Fee Sensitivity	0.24	Non-Int Share	6.7	6.2
α_{low}^d	Spread Sensitivity	0.99	Deposit Spread	85.0	86.3
$\bar{q}_{low}$	Demand Intercept	2.76	Deposit-Asset Ratio	72	63
ξ_{low}	Deposit Shifter	1.45	Deposit Beta	0.32	0.36
ϕ_{low}^π	Dividend Adjustment	4.69	Dividend Yield	0.58	0.89
ϕ_{low}^ℓ	Loan Cost	0.08	Loan Marginal Cost	0.45	1.34
α_{low}^n	Account Fee Sensitivity	1.20	Non-Int Share	1.3	1.3

Notes: Model parameters are set for a model in quarterly frequency. Data and Model moments are quoted in annualized terms.

We also estimate the cumulative deposit rate pass-through over eight quarters from a 100 basis point monetary policy shock. Specifically, we measure pass-through using a procedure that is consistent with the empirical specification in Section 3.2, ensuring that the results are directly comparable. The model-implied deposit rate pass-through is 43.8 and 53.0 basis points for the high and low NII banks, respectively. This captures the main empirical finding that the high NII bank exhibits lower rate pass-through. Figure 5 further shows that these magnitudes fall within the range of empirical estimates. In addition, we compare the model-implied pass-through of monetary policy shocks to fees with its empirical counterpart and find that both are quantitatively small in terms of percentage change.

FIGURE 5: DEPOSIT RATE PASS-THROUGH (MODEL AND EMPIRICAL MOMENTS)



Notes: Cumulative change in deposit rates based upon local projection estimates in Figure 2 and an internally-run model local projection in response to a 100 basis point monetary policy shock.

The model matches the empirical patterns well in Appendix Figure A.20, most noticeably the positive relationship between deposit spreads and monetary policy (left panel) and the negative relationship between monetary policy and non-interest income shares (right panel). The former arises from the model's demand system, whereby depositors become less elastic in response to an increase in the monetary policy rate. The latter is consistent with the evidence in Figure 1 and the broader finding that non-interest income is relatively more important in low interest rate environments—this pattern reflects the fact that account fees are less cyclical and do not vary directly with monetary policy. As a result, when interest rates and deposit spreads are low, non-interest income accounts for a larger share of total bank income.

In Appendix Figure A.21, we estimate model-based local projections to compare the effects of monetary policy shocks on balance sheet quantities with the empirical estimates reported in the paper. Consistent with the rate pass-through results, the model generates qualitatively similar dynamics: in response to a 100 basis point monetary policy shock, both loan and deposit quantities decline modestly and remain largely negative throughout the eight quarter horizon. We also examine how deposits adjust across the intensive and extensive margins. Appendix Figure A.22 decomposes the response of deposits into changes in the number of accounts (the extensive margin) and deposits per account (the intensive margin). Most of the deposit outflow occurs through the intensive margin rather than account closures.

5.3 Counterfactual Analysis

This section leverages novel features of the quantitative model—specifically, its incorporation of bank non-interest income and account demand—to conduct counterfactual analyses of the effects on credit supply, bank risk, and deposit pricing.

Bank Income and Monetary Policy Transmission. Section 3 shows that high non-interest income banks exhibit lower pass-through of monetary policy shocks to deposit pricing. However, the reduced form estimates do not imply a causal relationship. Indeed, Section 4 introduces an illustrative model in which a third factor—deposit market power—jointly determines both non-interest income shares and deposit pricing behavior. Using the quantitative model, we now examine whether exogenous changes in bank non-interest income itself can affect monetary policy transmission. To accomplish this, we introduce an anticipated and persistent stream of income $\bar{n}$ into the bank’s budget constraint. This can be interpreted as net profit from some other business line of the bank—such as investment banking operations—which is orthogonal to the deposit-loan business of the bank.

Table 8 reports key model moments for varying levels of the income endowment, measured as a share of total bank income in the baseline model.²⁸ Both panels show that exogenous increases in bank income reduce deposit rate pass-through, although the quantitative effects are modest. For example, even with a 100% income endowment, the 8-quarter cumulative deposit rate pass-through declines by only 0.8 and 1.6 basis points for the high and low NII bank, respectively.

Because average deposit spreads are also higher in this scenario, the deposit rate pass-through is relatively more muted, leading to a smaller contraction in lending. Thus, higher income dampens the transmission of monetary policy to lending. We also observe that bank risk (as proxied by the z-score) decreases as banks receive a larger endowment. Taken together, these results suggest that most of the observed co-movement between bank non-interest income and deposit rate pass-through is driven by other sources, such as market power and other market frictions, rather than by the endowment effect of non-interest income itself.

²⁸For example, if average bank income in the simulated baseline model is 1, then a value of 10% corresponds to $\bar{n} = 0.1$ in the counterfactual model.

TABLE 8: EXOGENOUS CHANGES IN BANK INCOME

		-20%	0%	20%	50%	100%	200%
Panel A: High NII Bank							
Deposit Rate Pass-Through	(bps)	44.6	43.8	43.4	43.2	43.0	42.9
Loan Quantity	(pp)	-0.46	-0.4	-0.36	-0.34	-0.32	-0.31
Deposit Spread	(bps)	75.0	103.8	111.3	136.7	146.4	153.7
Z-score	(int)	-1.2	17.7	34.3	50.5	61.5	69.8
Panel B: Low NII Bank							
Deposit Rate Pass-Through	(bps)	53.8	52.9	52.3	51.8	51.3	48.4
Loan Quantity	(pp)	-0.26	-0.24	-0.23	-0.22	-0.21	-0.15
Deposit Spread	(bps)	79.8	86.3	91.1	96.0	99.5	124.6
Z-score	(int)	25.8	31.7	36.5	40.9	44.3	51.6

Notes: Deposit rate and loan pass-through are measured as the cumulative rate pass-through over an 8-quarter period in response to a 100 basis point monetary policy shock, and it is measured in basis points and percentage points, respectively. Each column corresponds to an exogenous income endowment to the bank, where the size of the endowment is measured as a percentage of net income in the simulated baseline model for each bank.

Fee Caps. In the second counterfactual, we impose a fee constraint, $f_i \leq \bar{f}_i$, to examine its effect on bank pricing, leverage, and risk. This exercise is motivated by the fact that price caps have frequently been proposed and implemented for a variety of banking products.²⁹ We calibrate the cap $\bar{f}_i$ using percentiles of the simulated fee distribution from the baseline, unconstrained model. For example, $\bar{f}_i = 90\%$ corresponds to a cap that would bind in 10% of cases in the baseline model, absent any endogenous bank response.

Figure 6 traces the effects of increasingly binding fee caps for each bank type, moving from the unconstrained equilibrium (100% on the horizontal axis) toward tighter restrictions on the left.³⁰ As fee caps become more restrictive, banks increase deposit betas and lower deposit spreads. Although banks might be expected to offset lost fee income by extracting more through interest margins, this substitution effect is dominated by a leverage motive. In particular, lower fee income increases the likelihood that banks enter low networth states in which they have high demand for external funding (i.e., deposits), inducing them to offer lower deposit spreads. As a result, banks operate with higher leverage on average, increasing risk as reflected in lower z-scores.³¹

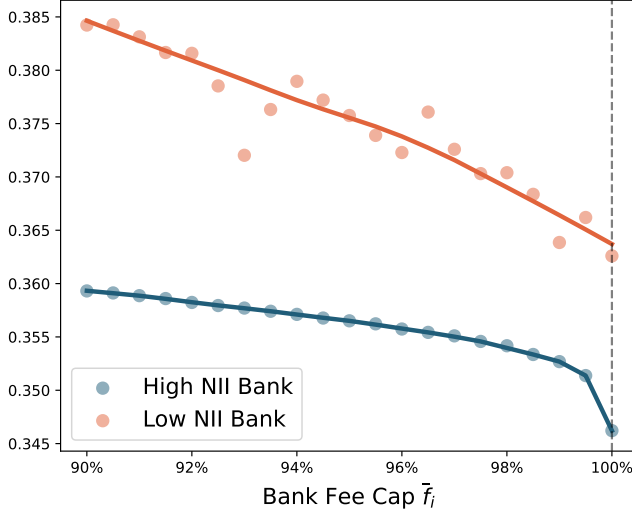
²⁹In the US, examples include Regulation Q (1933) for deposit rates, the Credit CARD Act of 2009 for credit card fees, and the Durbin Amendment (2011) for interchange fees on debit card transactions.

³⁰Appendix Section D provides additional details on the solution method and the construction of counterfactual outcomes.

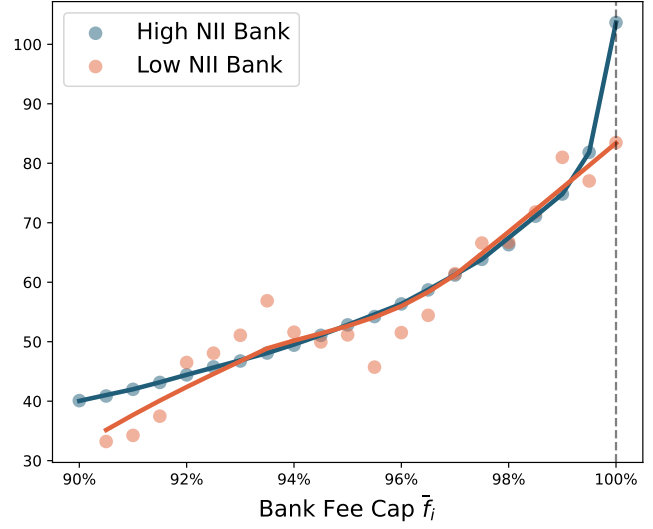
³¹Appendix Table A.3 reports the full set of counterfactual model moments. Note that model-implied z-scores can become negative in low networth states. The construction and interpretation of these values are discussed in Appendix Section E.

FIGURE 6: BANK MOMENTS GIVEN CAP ON FEE PRICING

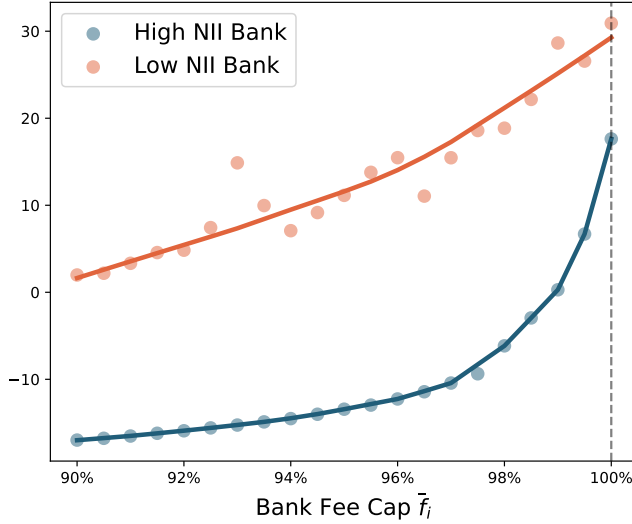
(a) DEPOSIT BETA



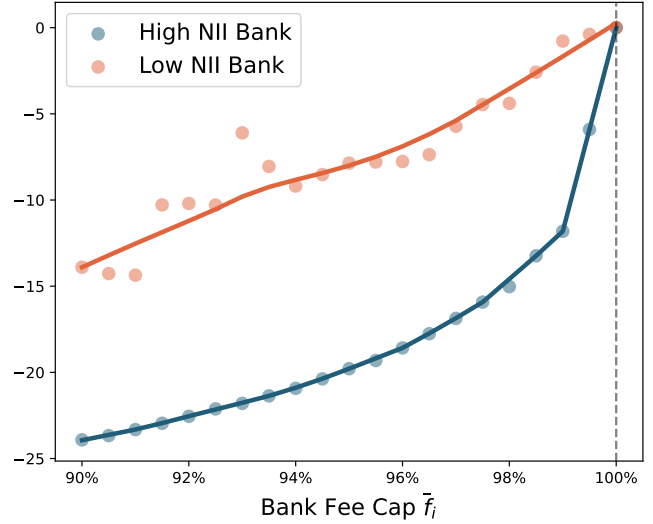
(b) DEPOSIT SPREAD



(c) BANK Z-SCORE



(d) % CHANGE LENDING



Notes: Fee caps $\bar{f}_i$ correspond to percentiles of the distribution of fees when the model has no fee constraints. Thus, lower values of $\bar{f}_i$ correspond to increasingly binding levels of the constraint.

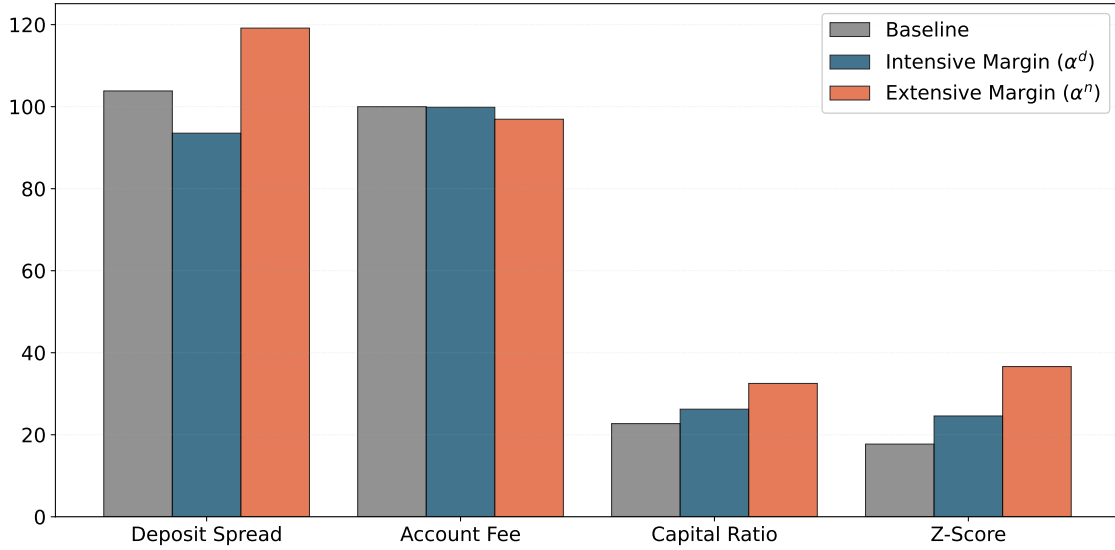
These findings underscore an important tension between consumer protection and financial stability. Although depositors benefit from lower deposit account prices, these gains arise through a riskier banking system. While our framework does not model systemic risk or broader spillovers from financial fragility, the results suggest that fee caps could increase both the frequency and severity of such costs.

Deposit Market Power and Bank Risk. Last, we examine how changes in deposit market power affect deposit pricing and bank risk, and whether the *margin* of demand matters. We model this by separately varying the price sensitivity parameters α_i^d and α_i^n . An increase in α_i^d corresponds to stronger competition along the intensive margin, where banks may lose deposits while retaining accounts, whereas an increase in α_i^n corresponds to stronger competition along the extensive margin, where banks may lose accounts altogether. To make the counterfactual changes comparable, each parameter is calibrated to generate a 10% decline in deposit demand in the baseline model, holding prices constant.³²

The general finding is that an increase in deposit competition—whether arising through the intensive or extensive margin—leads to lower bank risk, although the magnitude and direction of the effects vary. Figure 7 presents outcomes for the High NII bank. When spread sensitivity α_i^d increases, the bank responds by lowering deposit spreads, reflecting a lower markup under more elastic demand. Because deposit funding becomes effectively more expensive, the bank chooses lower average leverage. The result is a higher capital ratio and a higher z-score, indicating less bank risk. A similar mechanism operates when fee sensitivity α_i^n increases: the bank reduces fee markups and also widens deposit spreads to reduce leverage. This scenario produces larger effects on bank risk than the intensive margin scenario because the extensive margin directly affects both interest income and fee income.

³²For the intensive margin exercise, $(\alpha_{High}^d, \alpha_{Low}^d)$ increases from (0.62, .099) to (0.67, 1.09). For the extensive margin exercise, $(\alpha_{High}^n, \alpha_{Low}^n)$ increases from (0.25, 1.20) to (0.27, 1.24).

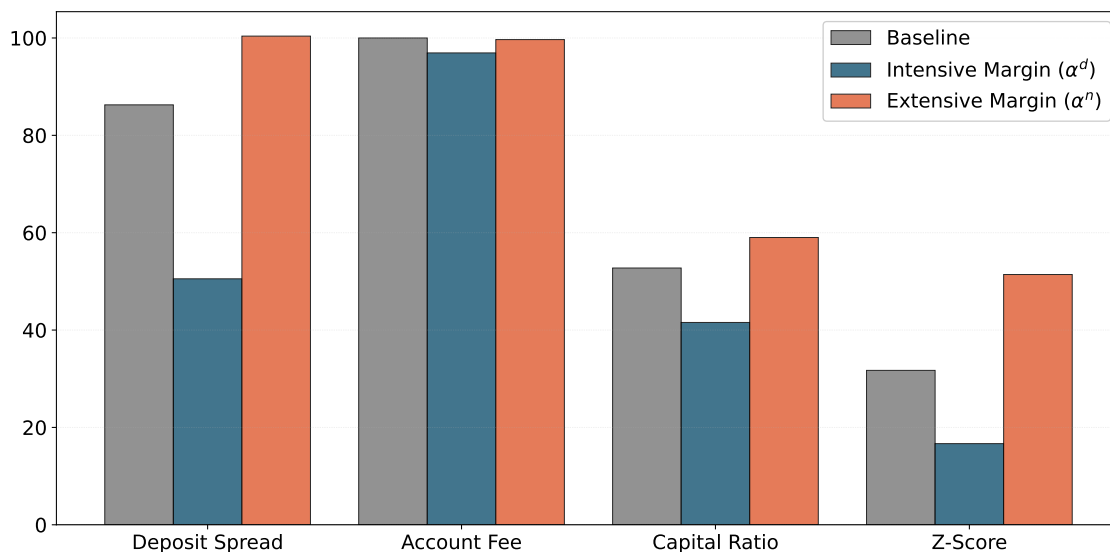
FIGURE 7: MODEL MOMENTS UNDER CHANGING MARKET POWER (HIGH NII BANK)



Notes: This figure plots model moments for the High NII bank in the baseline model as well as two counterfactual scenarios: first, an increase in deposit spread sensitivity α^d (the intensive margin) and, second, an increase in account fee sensitivity α^n (the extensive margin). For each counterfactual scenario, the change in the price sensitivity parameter corresponds to a 10% drop in total deposit demand in the baseline model, holding prices constant. Account fees are normalized to 100 in the baseline model.

Results for the Low NII bank in Figure 8 show a different outcome in the intensive margin scenario: higher spread sensitivity α_i^d increases bank risk. This divergence is driven by the cross-subsidizing feature of account pricing, where the optimal fee is given by $f_i = \phi_i^n + \frac{N_i}{\partial N_i / \partial f_i} - \frac{E[m'_i]d_i^2}{\alpha_i^d}$. For the Low NII bank, the relative importance of the intensive margin $\left(\frac{E[m'_i]d_i^2}{\alpha_i^d}\right)$ actually increases in this scenario, placing downward pressure on fees and compresses already low amounts of fee income. As a result, the bank is more likely to end up in low networth states in which it optimally increases leverage. This effect dominates such that the bank has a lower average capital ratio and z-score.

FIGURE 8: MODEL MOMENTS UNDER CHANGING MARKET POWER (LOW NII BANK)



Notes: This figure plots model moments for the Low NII bank in the baseline model as well as two counterfactual scenarios: first, an increase in deposit spread sensitivity α^d (the intensive margin) and, second, an increase in account fee sensitivity α^n (the extensive margin). For each counterfactual scenario, the change in the price sensitivity parameter corresponds to a 10% drop in total deposit demand in the baseline model, holding prices constant. Account fees are normalized to 100 in the baseline model.

While we generally find that competition leads banks to deleverage and become safer, the result is not universal. Instead, the effects vary across banks and scenarios. In a related modeling framework, Corbae and D’Erasmus (2021) find that competition reduces bank failure rates, although their analysis focuses on lending market power. Similar to our findings, Whited et al. (2021) document ambiguous effects of competition on bank risk; specifically, competition reduces risk only in low-rate environments.³³ More broadly, our results reinforce the view that the relationship between competition and bank stability is not monotonic (see, for example, Martinez-Miera and Repullo 2010). From a policy perspective, these findings highlight the need for supervisory and regulatory frameworks that account for heterogeneity in banks’ risk responses to competitive pressures.

6 Conclusion

This paper examines the relationship between deposit non-interest income, the pricing of deposit accounts through rates and fees, and the transmission of monetary policy. We show empirically that

³³Our results are qualitatively unchanged across low- and high-interest-rate environments in the counterfactual analysis.

deposit non-interest income is strongly correlated with bank pricing power and lower deposit rate pass-through of monetary policy shocks. To rationalize these findings, we develop an illustrative model in which differences in deposit market power generate the observed relationship between non-interest income and deposit pricing. We then develop a larger, quantitative model of the US banking system to conduct counterfactual analysis. The quantitative model reinforces the central role of deposit market power in explaining the empirical correlations while also highlighting the importance of non-interest income for bank credit supply, balance sheet risk, and deposit pricing.

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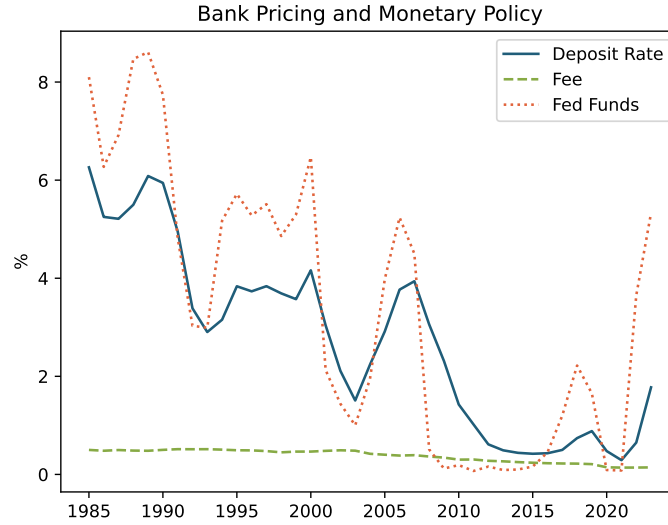
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A Additional Figures and Tables

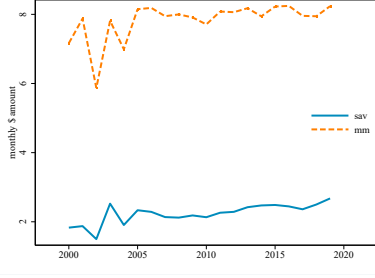
FIGURE A.1: DEPOSIT RATES AND FEES OVER TIME



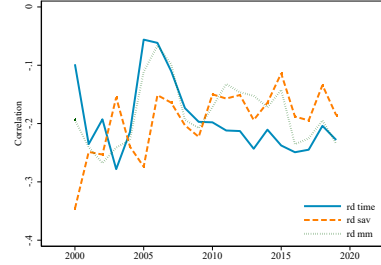
Notes: Deposit rate computed as the ratio of total deposit interest expense to total deposits. Fee rate computed as the ratio of service charge income on deposit accounts to total deposits. Numbers are annualized by using comprehensive Q4 expense and income for each fiscal year. Source: FDIC.

FIGURE A.2: BRANCH-LEVEL FEES OVER TIME

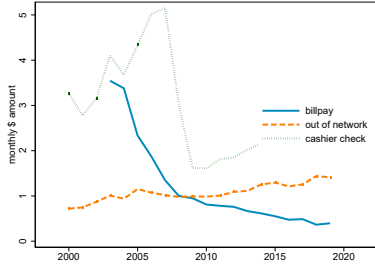
(a) AVERAGE MONTHLY CHARGES



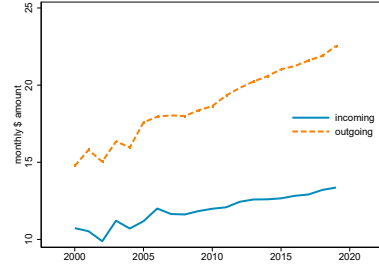
(b) CORRELATION OF MONTHLY CHARGES AND RATES



(c) AVERAGE CHARGES TRANSACTIONS BRANCH

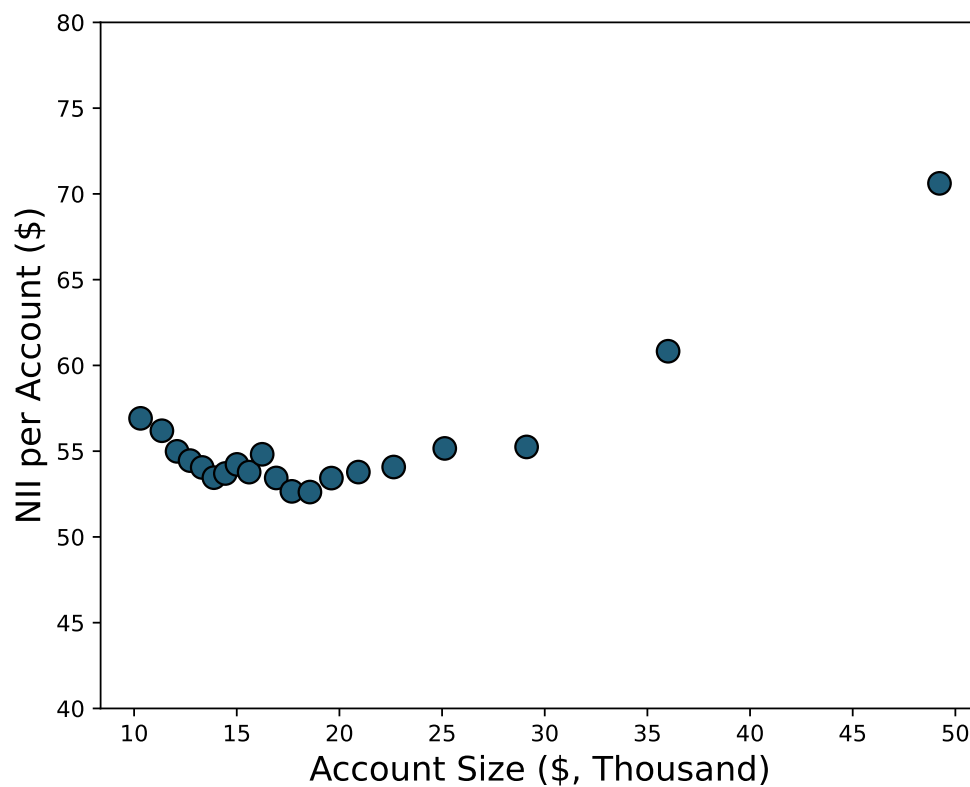


(d) AVERAGE CHARGES WIRE TRANSFER DOMESTIC



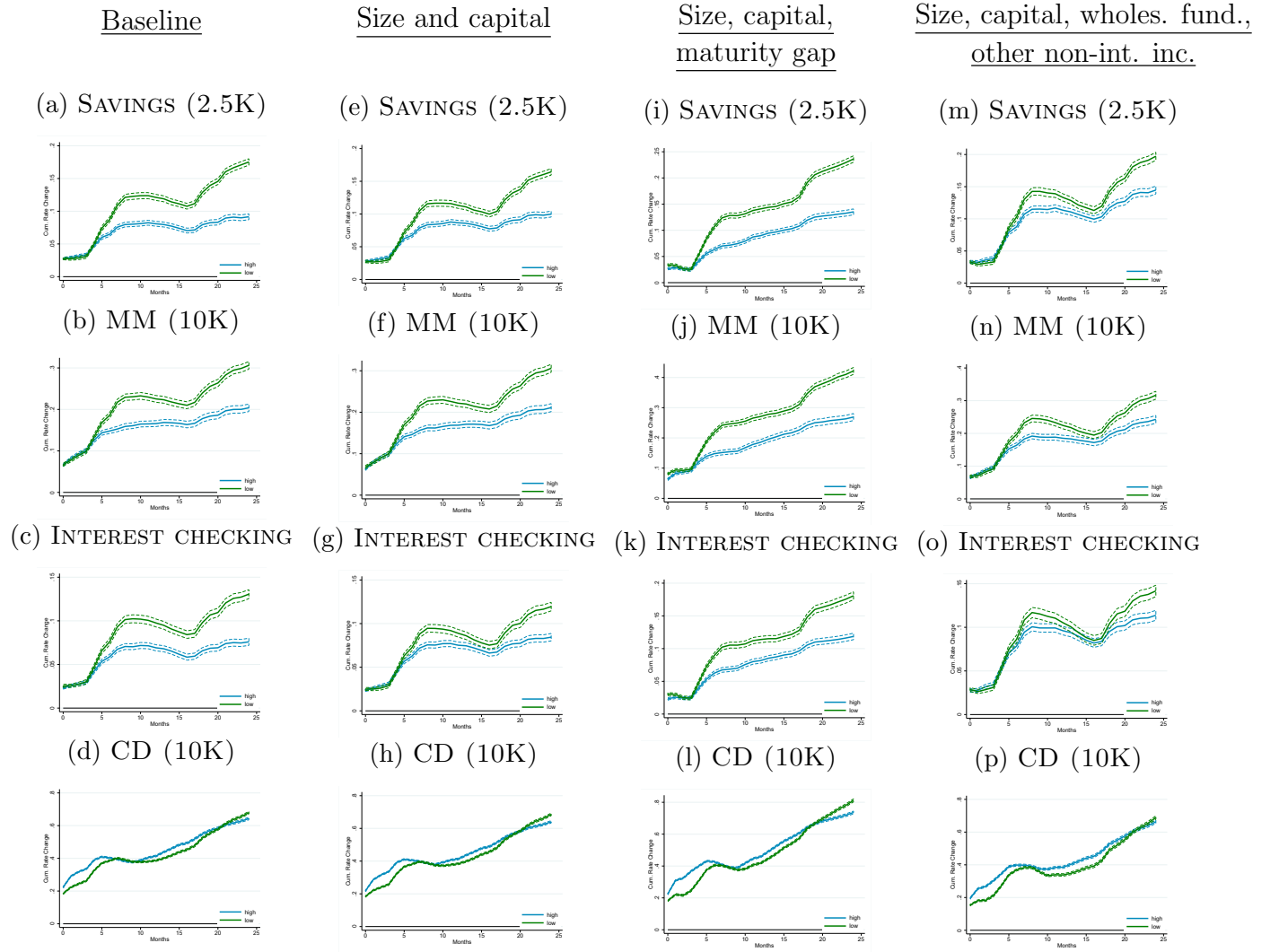
Notes: Panel A shows average monthly charges for savings and money market accounts over time. Panel B shows the correlation coefficient of monthly charges and deposit rates for each point in time. Panel C shows average charges for transactions (billpay, out-of-network terminal use, and cashier checks) over time. Panel D shows averages of incoming and outgoing wire transfer charges over time. Converted to yearly frequency (average). Source: RateWatch.

FIGURE A.3: DEPOSITOR NII-PER-ACCOUNT AND DEPOSIT ACCOUNT SIZE



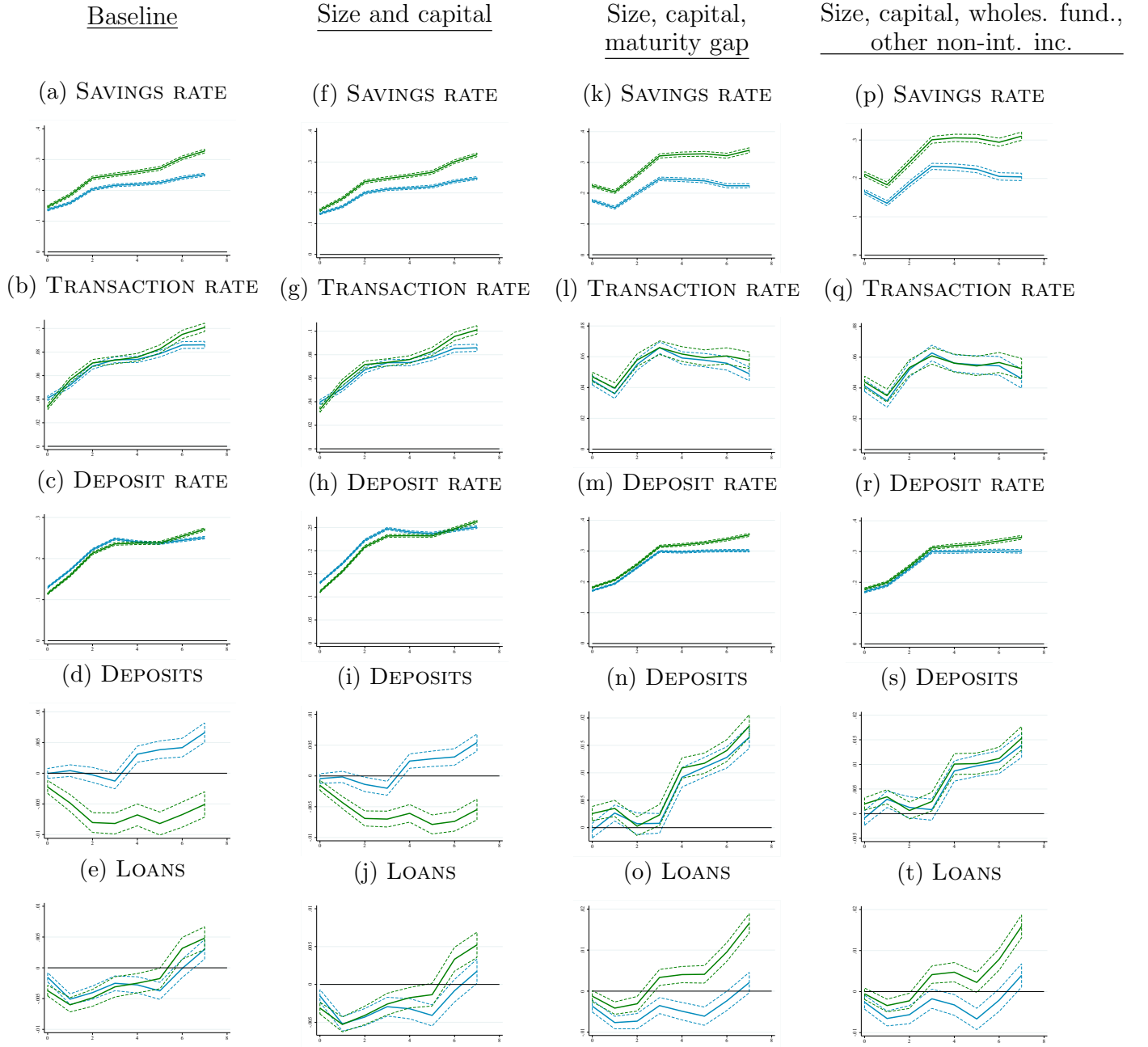
Notes: This chart creates a quantile-binned scatter plot between depositor non-interest income (NII) per account relative to the average deposit account size using bank-quarter observations. Source: Call Report.

FIGURE A.4: LOCAL PROJECTIONS OF DEPOSIT RATES TO MONETARY POLICY SHOCK (WITH AND WITHOUT ADDITIONAL CONTROLS)



Notes: Impulse responses of deposit rates to a monetary policy shock for both high (blue) and low (green) shares of deposit non-interest income. Column (1) presents the baseline results without controls. Column (2) controls for size (log assets) and capitalization (equity ratio) in levels, as well as interacted with the monetary policy shock. Column (3) additionally controls for the maturity gap, and column (4) additionally controls for the wholesale funding ratio and other non-interest income in those specifications in levels, as well as interacted with the monetary policy shock. Horizon in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

FIGURE A.5: LOCAL PROJECTIONS OF BANK VARIABLES TO MONETARY POLICY SHOCK (WITH AND WITHOUT ADDITIONAL CONTROLS)

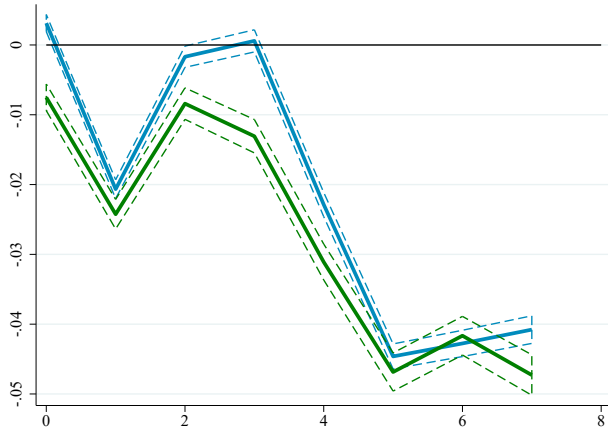


Notes: Impulse responses of deposit rates to a monetary policy shock for both high (blue) and low (green) shares of deposit non-interest income. Column (1) presents the baseline results without controls. Column (2) controls for size (log assets) and capitalization (equity ratio) in levels, as well as interacted with the monetary policy shock. Column (3) additionally controls for the maturity gap, and column (4) additionally controls for the wholesale funding ratio and other non-interest income in those specifications in levels, as well as interacted with the monetary policy shock. Horizon in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

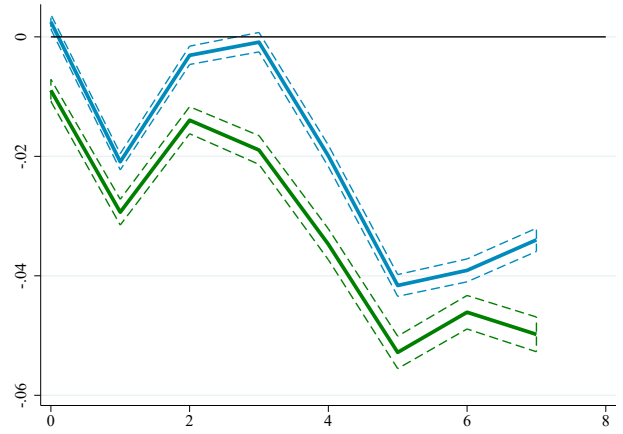
FIGURE A.6: BANK-LEVEL LOCAL PROJECTIONS OF FEES BY DEPOSIT NON-INTEREST INCOME SHARES

Panel A: Without controls

(a) LOG (FEE PRICE)

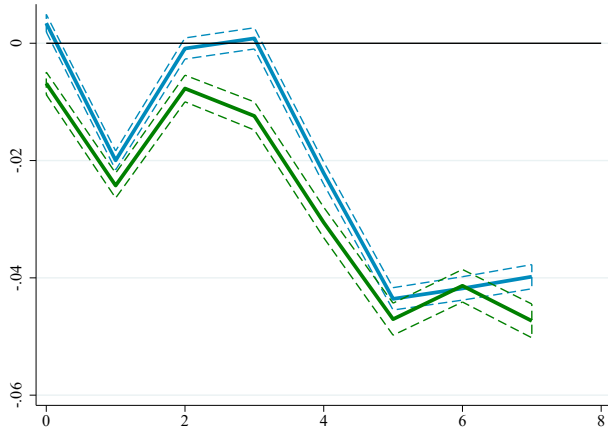


(b) LOG INCOME ON DEPOSIT SERVICES

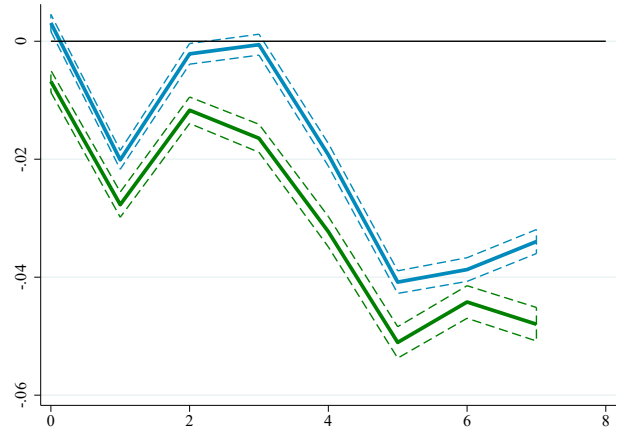


Panel B: With controls

(c) LOG (FEE PRICE)

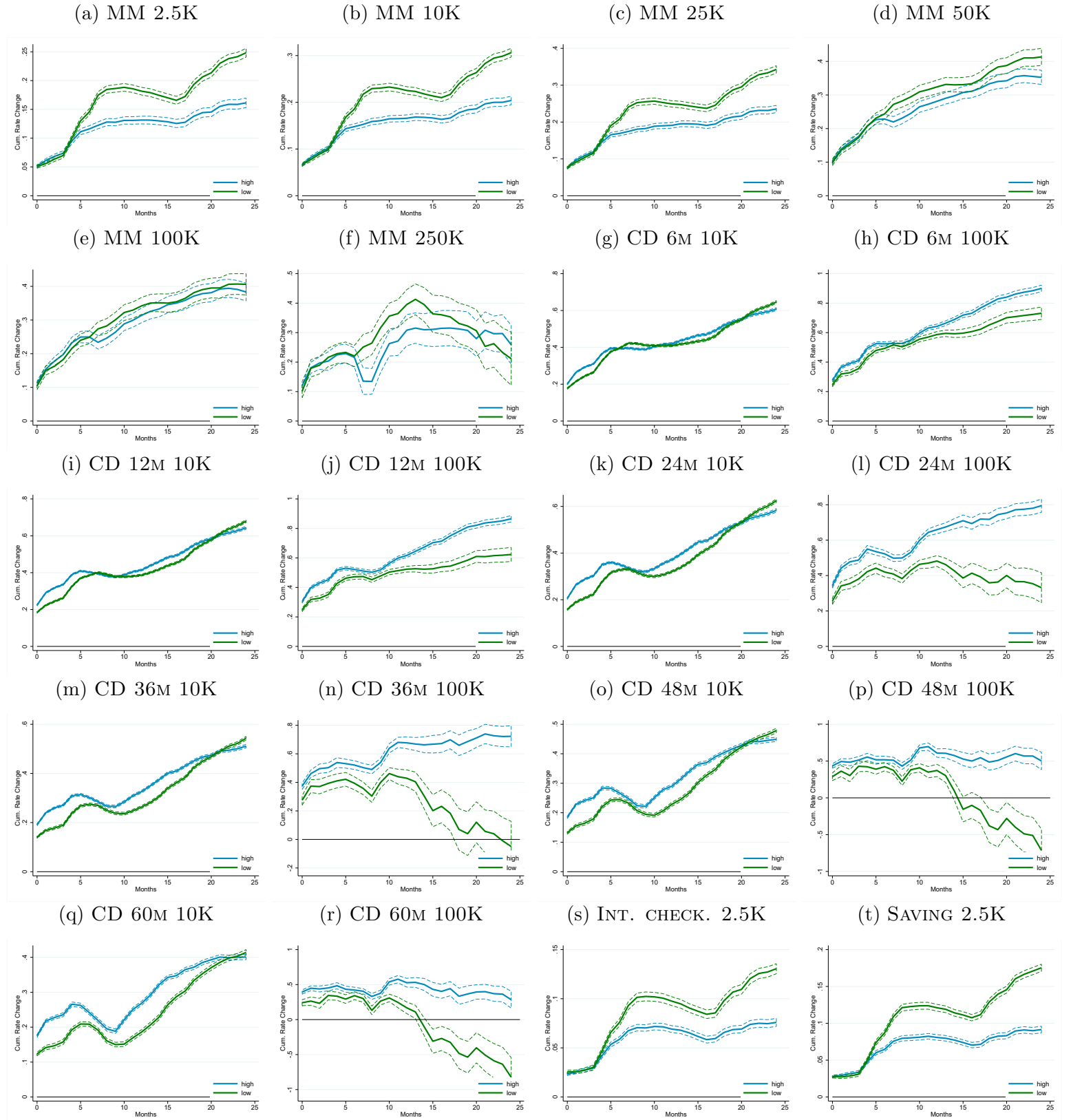


(d) LOG INCOME ON DEPOSIT SERVICES



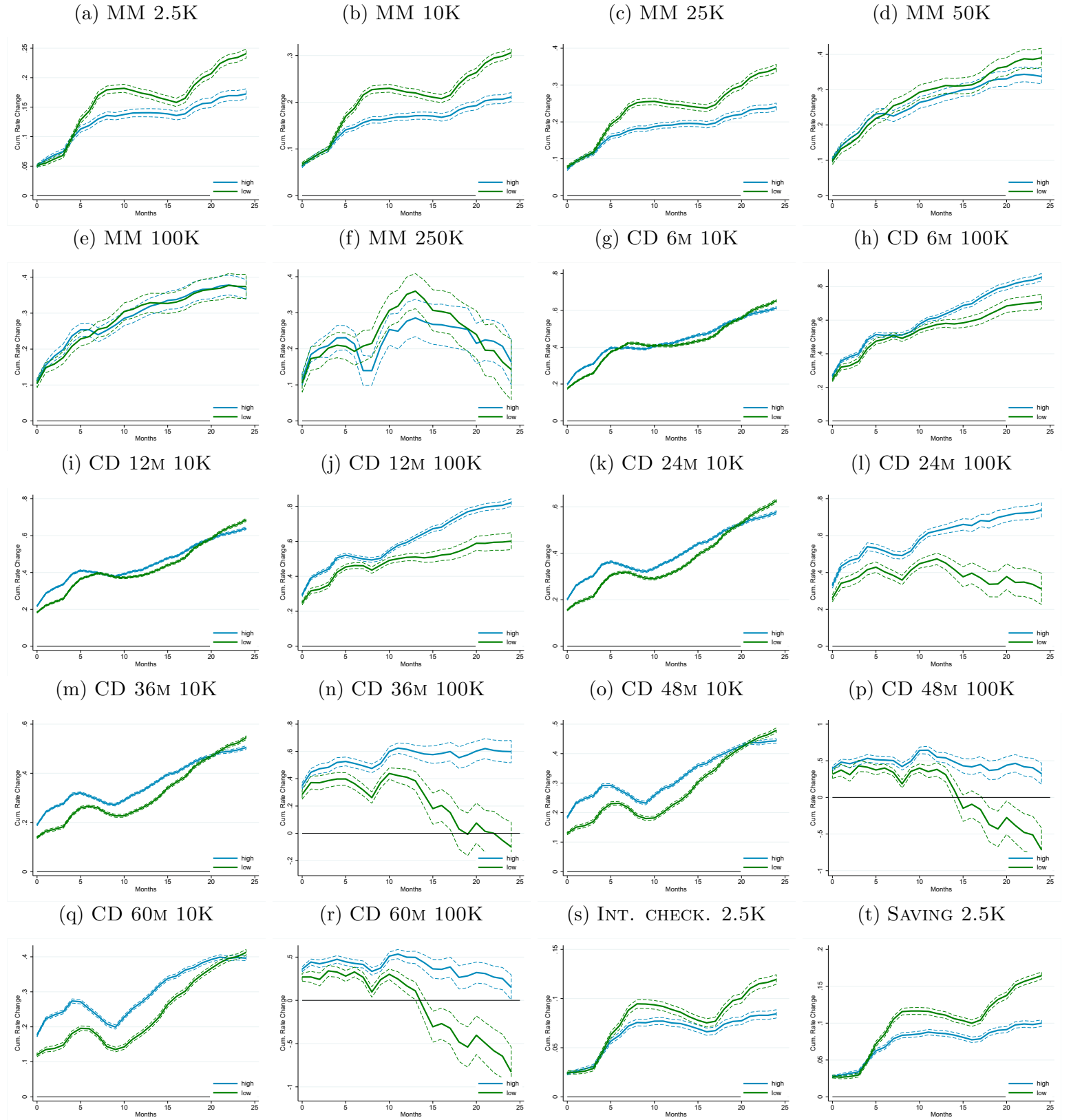
Notes: Impulse responses of log fee price and log income on deposit services to a monetary policy shock at both high (blue) and low (green) shares of deposit non-interest income. Panel B includes interactions of size and capitalization and the shock variable as controls. Horizon is in quarters. Standard errors are clustered by bank. 90% confidence intervals. Source: Call Report.

FIGURE A.7: LOCAL PROJECTIONS OF BRANCH-LEVEL DEPOSIT RATES TO MONETARY SHOCK BY DEPOSIT NON-INTEREST INCOME SHARE (WITHOUT CONTROLS)



Notes: Impulse responses of different deposit rates to a monetary policy shock at both high (blue) and low (green) shares of deposit non-interest income. Horizon is in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

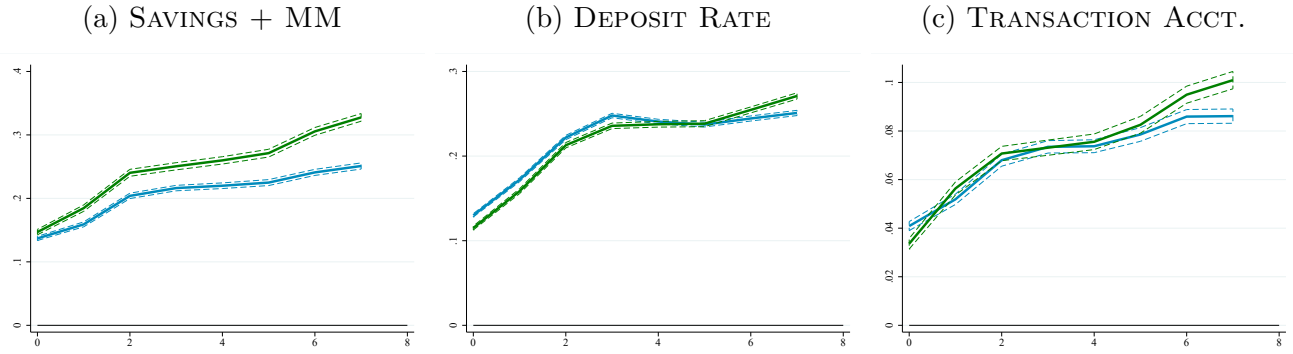
FIGURE A.8: LOCAL PROJECTIONS OF BRANCH-LEVEL DEPOSIT RATES TO MONETARY SHOCK BY DEPOSIT NON-INTEREST INCOME SHARE (WITH CONTROLS)



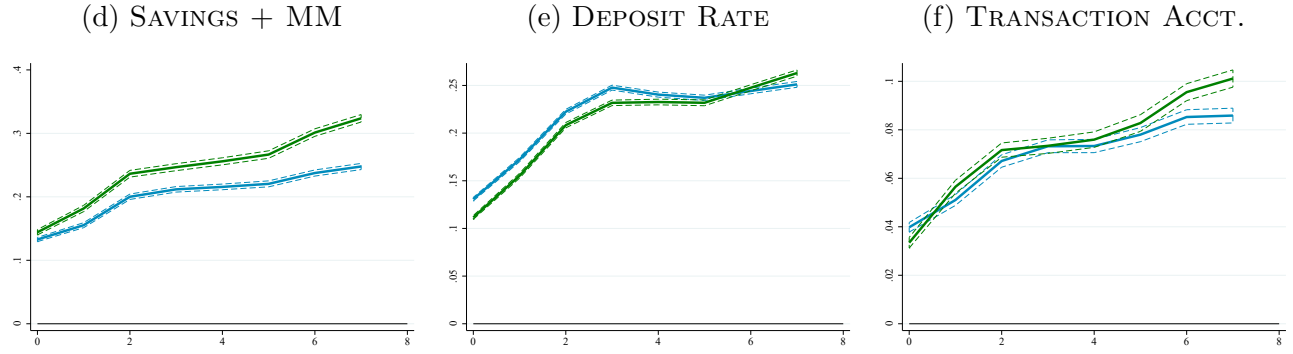
Notes: Impulse responses of different deposit rates to a monetary policy shock at both high (blue) and low (green) shares of deposit non-interest income. Regressions include interactions of size and capitalization and the shock variable as controls. Horizon is in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

FIGURE A.9: LOCAL PROJECTIONS OF DEPOSIT RATES TO MONETARY SHOCK BY DEPOSIT NON-INTEREST INCOME SHARE

Panel A: Without controls

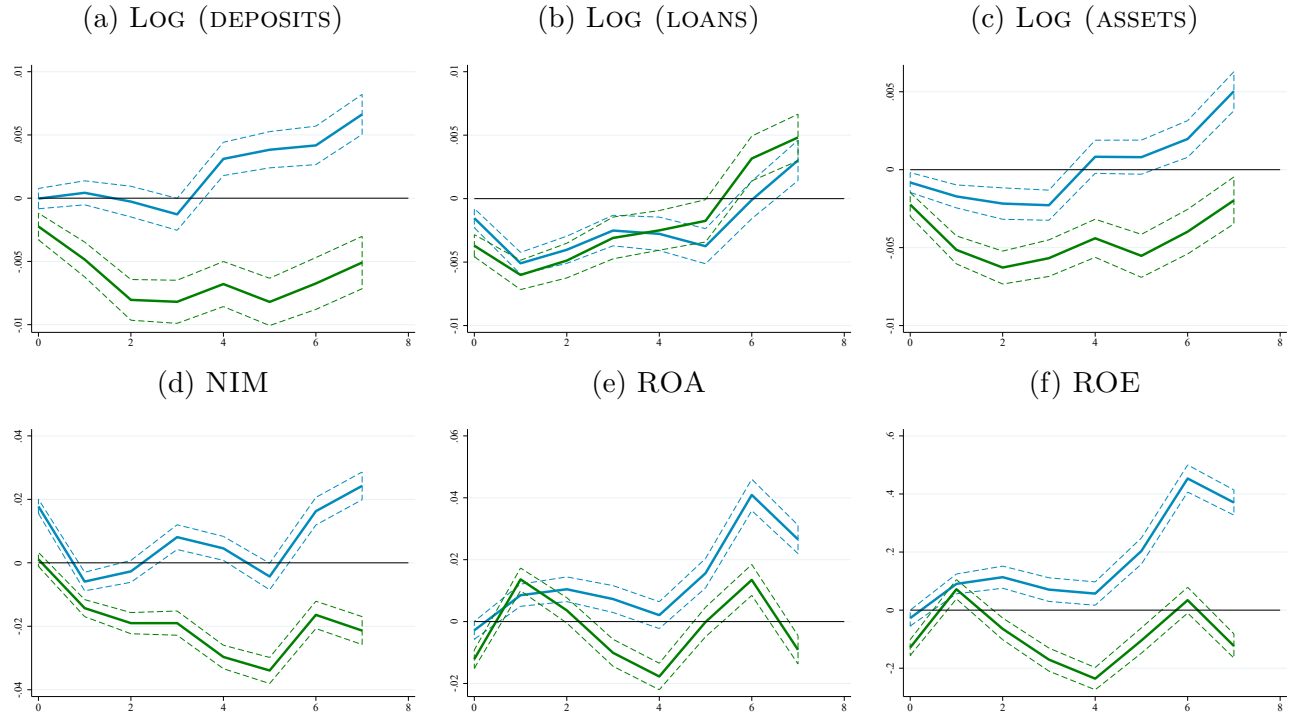


Panel B: With size and capitalization controls

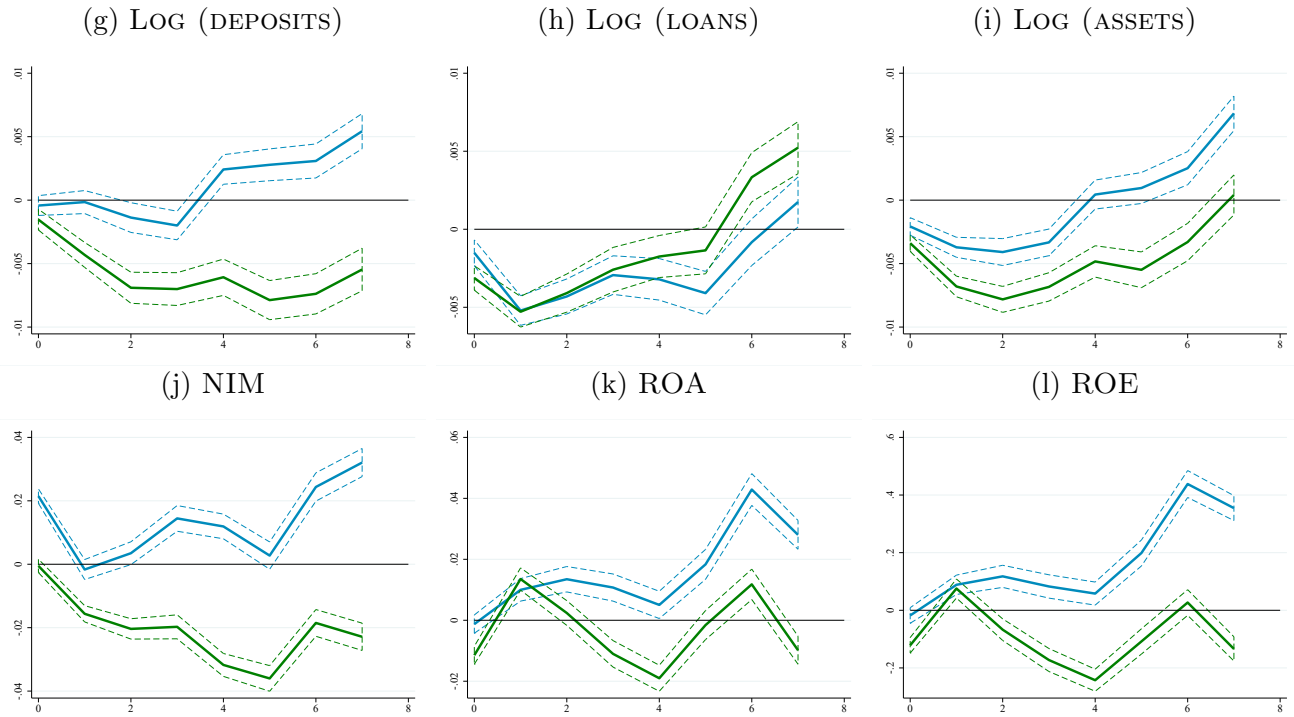


Notes: Impulse responses of different bank-level deposit rates to a monetary policy shock at both high (blue) and low (green) shares of deposit non-interest income. Panel B includes interactions of size and capitalization and the shock variable as controls. Horizon is in quarters. Standard errors are clustered by bank. 90% confidence intervals. Source: RateWatch.

FIGURE A.10: BANK-LEVEL LOCAL PROJECTIONS BY DEPOSIT NON-INTEREST INCOME SHARE
Panel A: Without controls

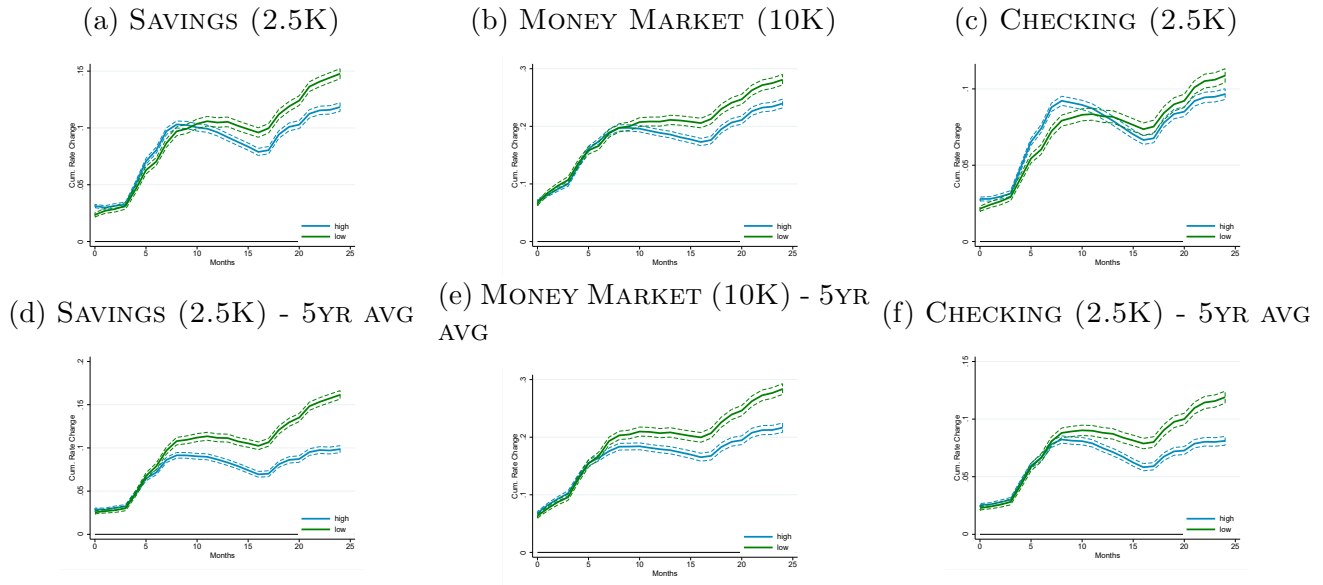


Panel B: With controls



Notes: Impulse responses of log deposits, log loans, log assets, net interest margin (NIM), return on assets (ROA), and return on equity (ROE) to a monetary policy shock at both high (blue) and low (green) shares of deposit non-interest income. Panel B includes interactions of size and capitalization and the shock. Horizon is in quarters. Standard errors are clustered by bank. 90% confidence intervals. Source: RateWatch.

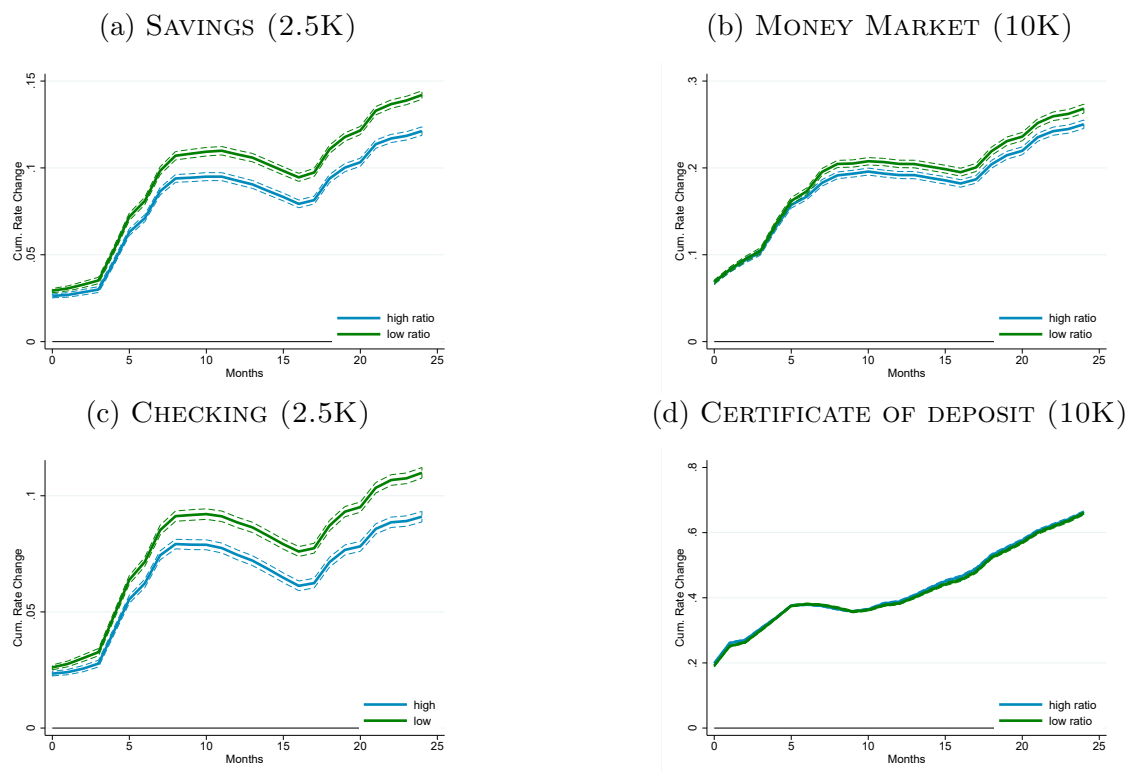
FIGURE A.11: FEE PRICE AS A PROXY FOR FEE DEPENDENCE



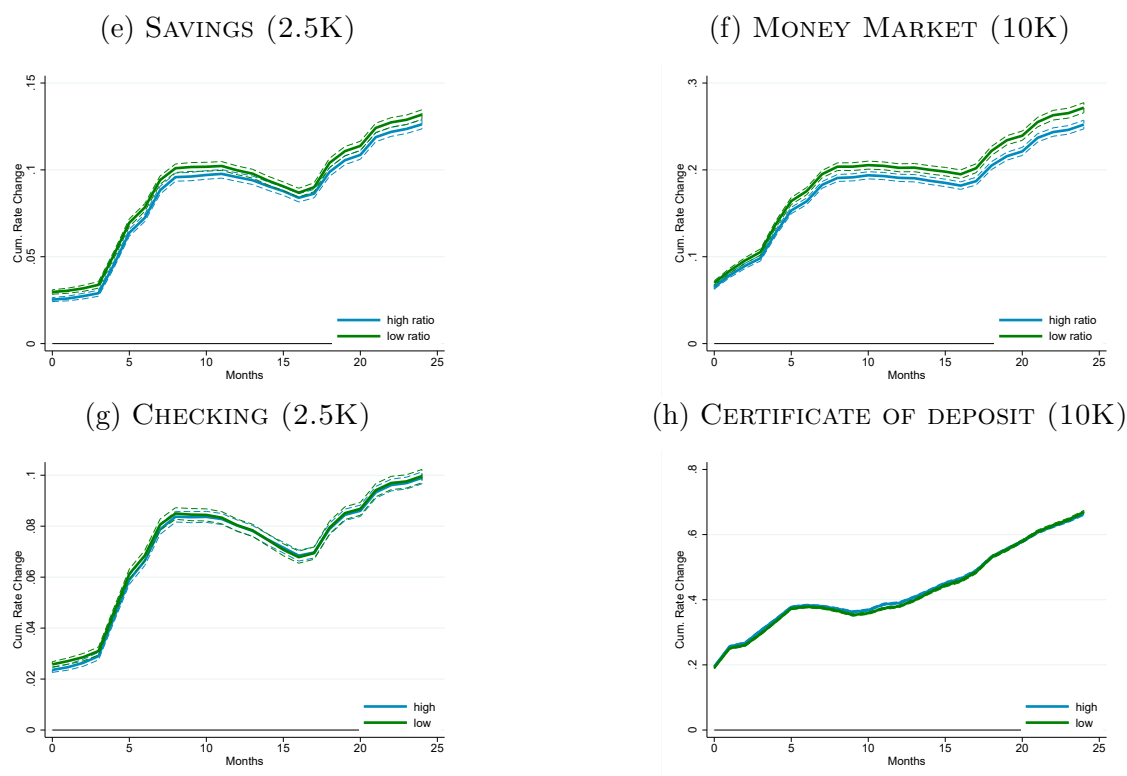
Notes: Impulse responses of deposit rates (savings accounts, money market accounts, checking accounts, and certificates of deposit) to a monetary policy shock at both high (blue) and low (green) fee prices (calculated as service charges on deposit accounts over deposits: $\text{iserchg}/\text{dep}$). Horizon is in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

FIGURE A.12: NUMBER OF BRANCHES AS A PROXY FOR LIQUIDITY SERVICES QUALITY

Panel A: Without controls



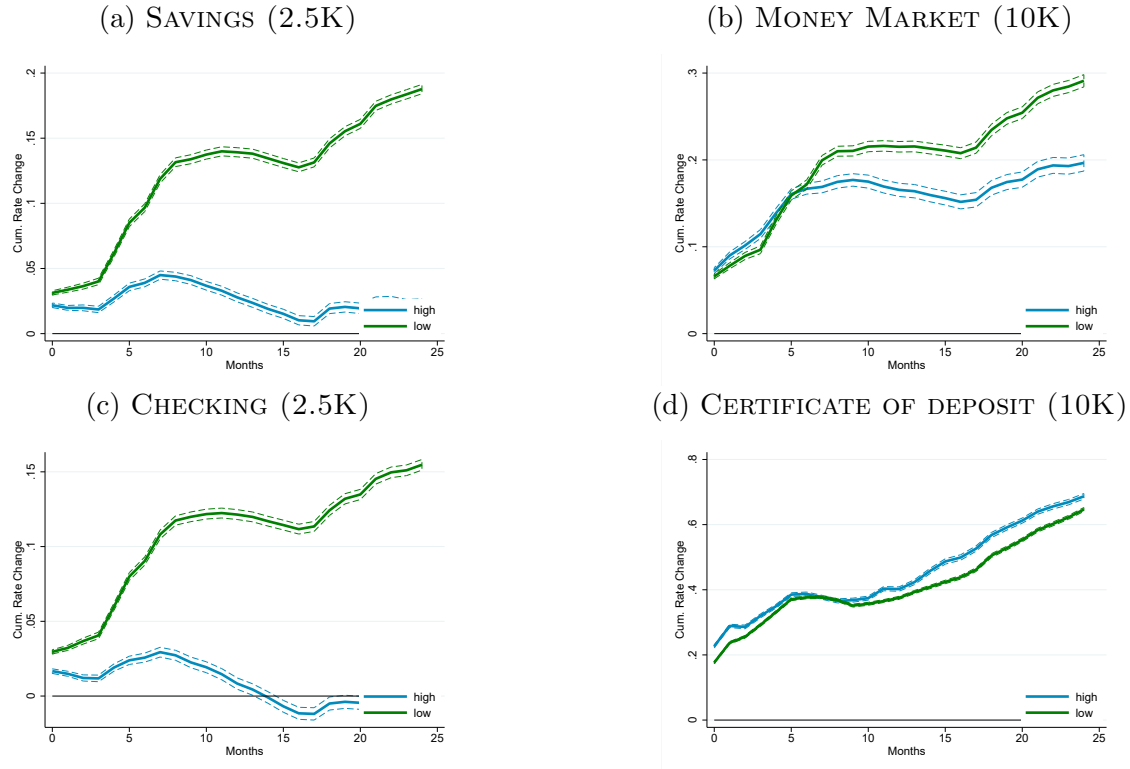
Panel B: With controls



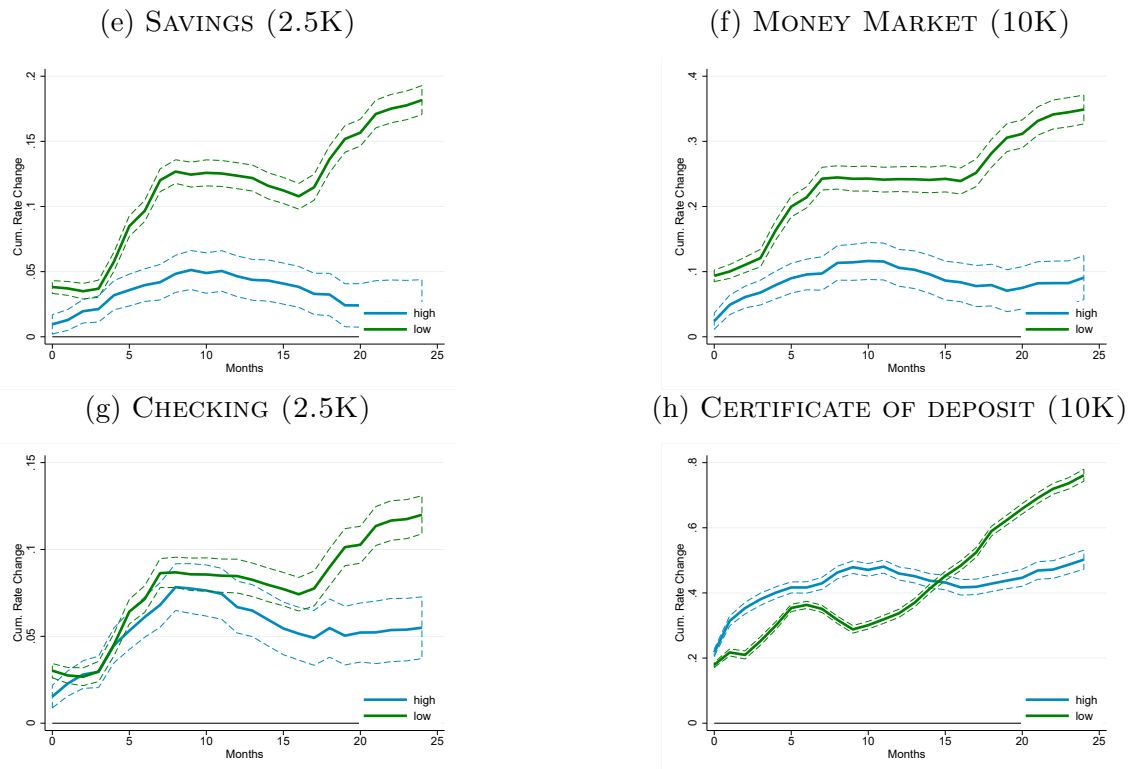
Notes: Impulse responses of deposit rates (savings accounts, money market accounts, checking accounts, and certificates of deposit) to a monetary policy shock at both high (blue) and low (green) number of branches. Panel B includes interactions of size and capitalization and the shock variable as controls. Horizon is in months. Standard errors are clustered by bank. 90% confidence intervals. ~~Source:~~ RateWatch.

FIGURE A.13: LOG NUMBER OF BRANCHES AS A PROXY FOR LIQUIDITY SERVICES QUALITY

Panel A: Without controls



Panel B: With size and capitalization controls



Notes: Impulse responses of deposit rates (savings accounts, money market accounts, checking accounts, and certificates of deposit) to a monetary policy shock at both high (blue) and low (green) log number of branches. Panel B includes interactions of size and capitalization and the shock variable as controls. Horizon is in months. Standard errors are clustered by bank. 90% confidence intervals. Source: RateWatch.

FIGURE A.14: LOCAL PROJECTIONS OF MM 10K RATE TO MONETARY POLICY

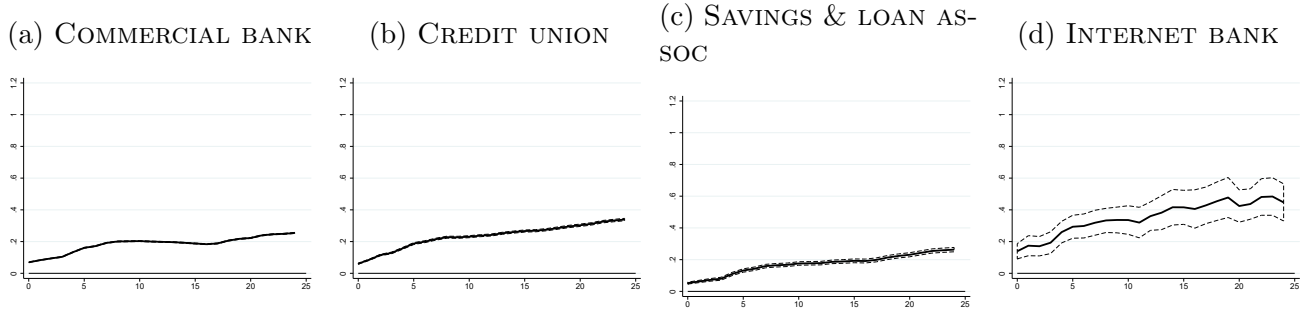


FIGURE A.15: LOCAL PROJECTIONS OF SAV 2.5K RATE TO MONETARY POLICY

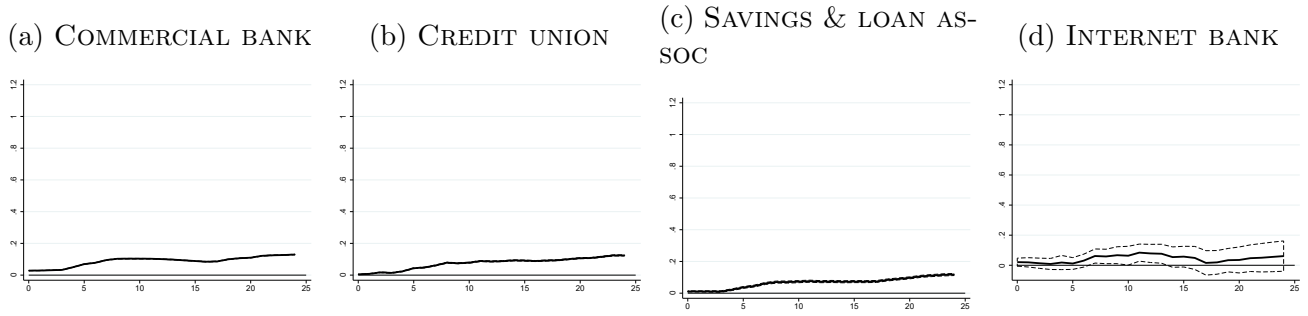


FIGURE A.16: LOCAL PROJECTIONS OF 12M CD 10K RATE TO MONETARY POLICY

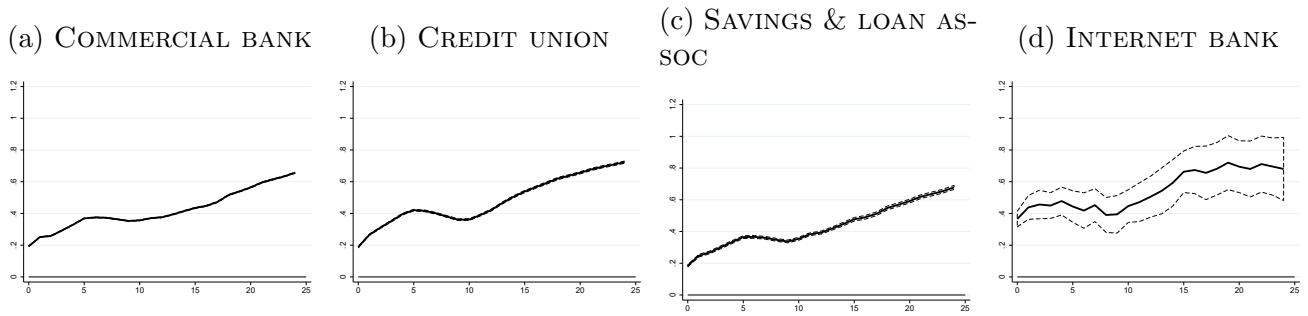


TABLE A.1: RATE PASS-THROUGH, DEPOSIT NON-INTEREST INCOME, AND CONCENTRATION

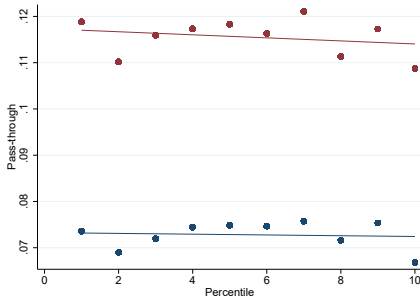
Panel A: Savings (2.5K)			
HHI $\times$ dFF	-0.000632 (-0.09)		0.00646 (1.06)
Deposit NII Share $\times$ dFF		-0.176* (-2.19)	-0.178* (-2.22)
N	1424723	1350218	1350218
R^2	0.0780	0.0812	0.0812
Panel B: Money Market (10K)			
HHI $\times$ dFF	-0.0502*** (-3.69)		-0.0350** (-2.97)
Deposit NII Share $\times$ dFF		-0.245 (-1.69)	-0.231 (-1.59)
N	1359216	1286576	1286576
R^2	0.0907	0.0916	0.0916
Panel C: Interest Checking (2.5K)			
HHI $\times$ dFF	0.000515 (0.08)		0.00735 (1.06)
Deposit NII Share $\times$ dFF		-0.113 (-1.64)	-0.115 (-1.66)
N	1370096	1297596	1297596
R^2	0.0666	0.0679	0.0679
Panel D: Certificate of Deposit (10K)			
HHI $\times$ dFF	-0.0496* (-2.28)		-0.0418 (-1.69)
Deposit NII Share $\times$ dFF		-0.107 (-0.37)	-0.0901 (-0.31)
N	1454042	1378535	1378535
R^2	0.229	0.231	0.231

Notes: This table reports estimated pass-through of policy rate changes to deposit rate changes for savings accounts (2.5K), money market accounts (10K), interest checking accounts (2.5K), and 12-month certificates of deposit (10K), conditional on county-level market concentration (HHI) and a bank's deposit non-interest income share, estimated separately and jointly. All specifications include branch and county fixed effects as well as macroeconomic controls. Standard errors are clustered at the county, bank, and time level. *, **, and *** indicate significance at the 5%, 1%, and 0.1% levels.

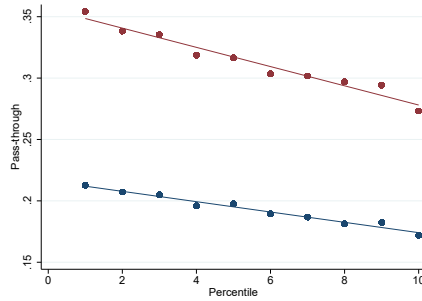
FIGURE A.17: BIN SCATTER DEPOSIT BETA BY DEPOSIT NON-INTEREST INCOME SHARE

Bank-level regressions

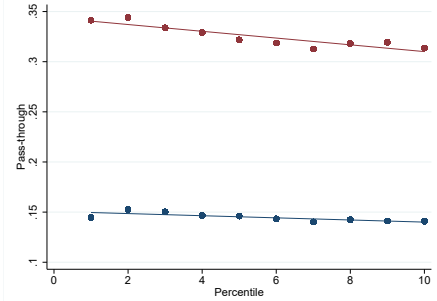
(a) TRANSACTION ACCOUNTS



(b) SAVINGS ACCOUNTS

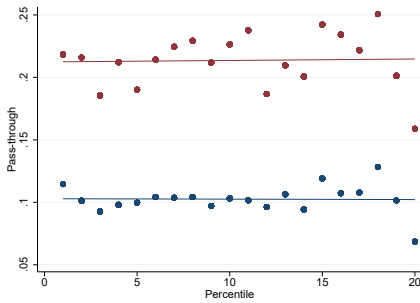


(c) DEPOSIT ACCOUNTS

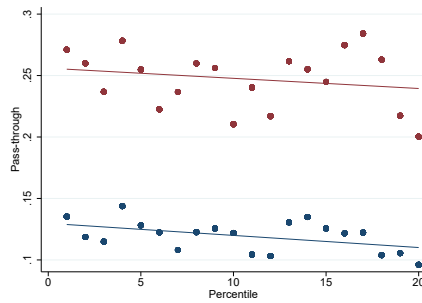


Branch-level regressions

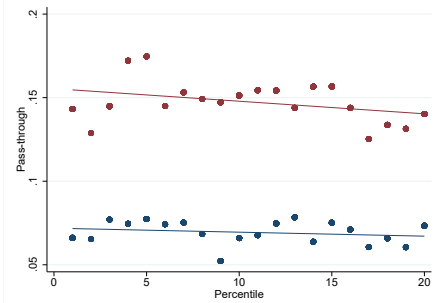
(d) MM 2.5K



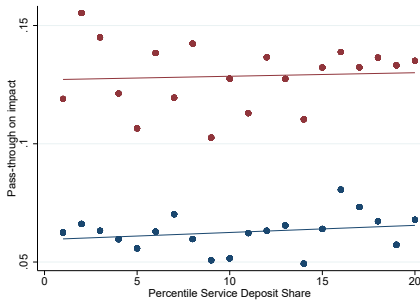
(e) MM 10K



(f) SAVINGS RATE 2.5K

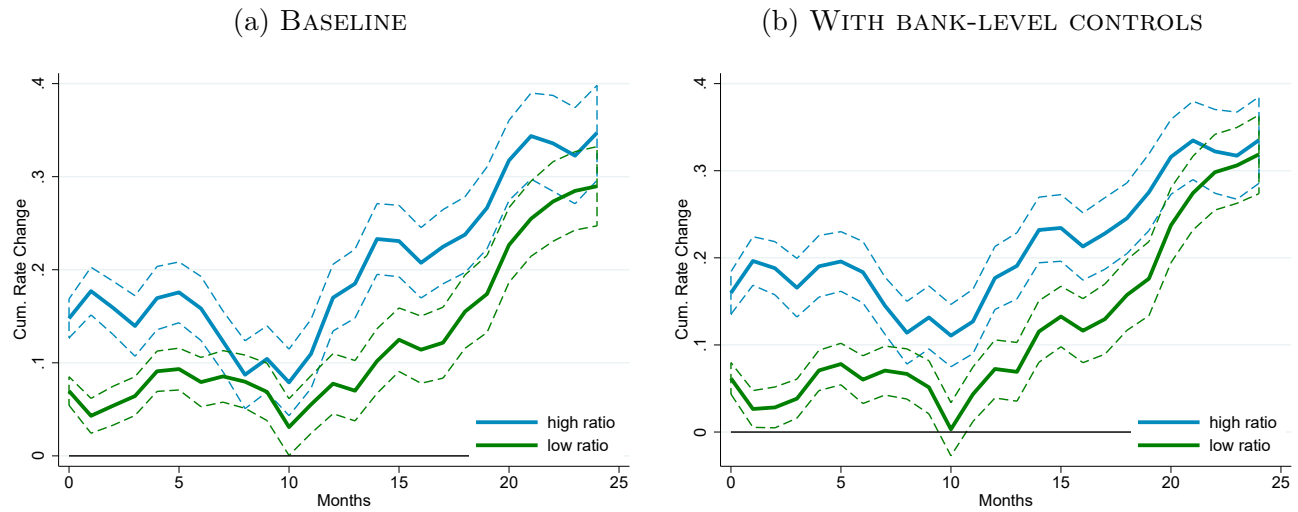


(g) INTEREST CHECKING



Notes: This figure shows binned scatter plots of deposit rate betas estimated at the bank level (upper panel) and branch level (lower panel) by service deposit share. The blue dots and line report the contemporaneous effects, and the red dots and line report cumulative effects over 12 months.

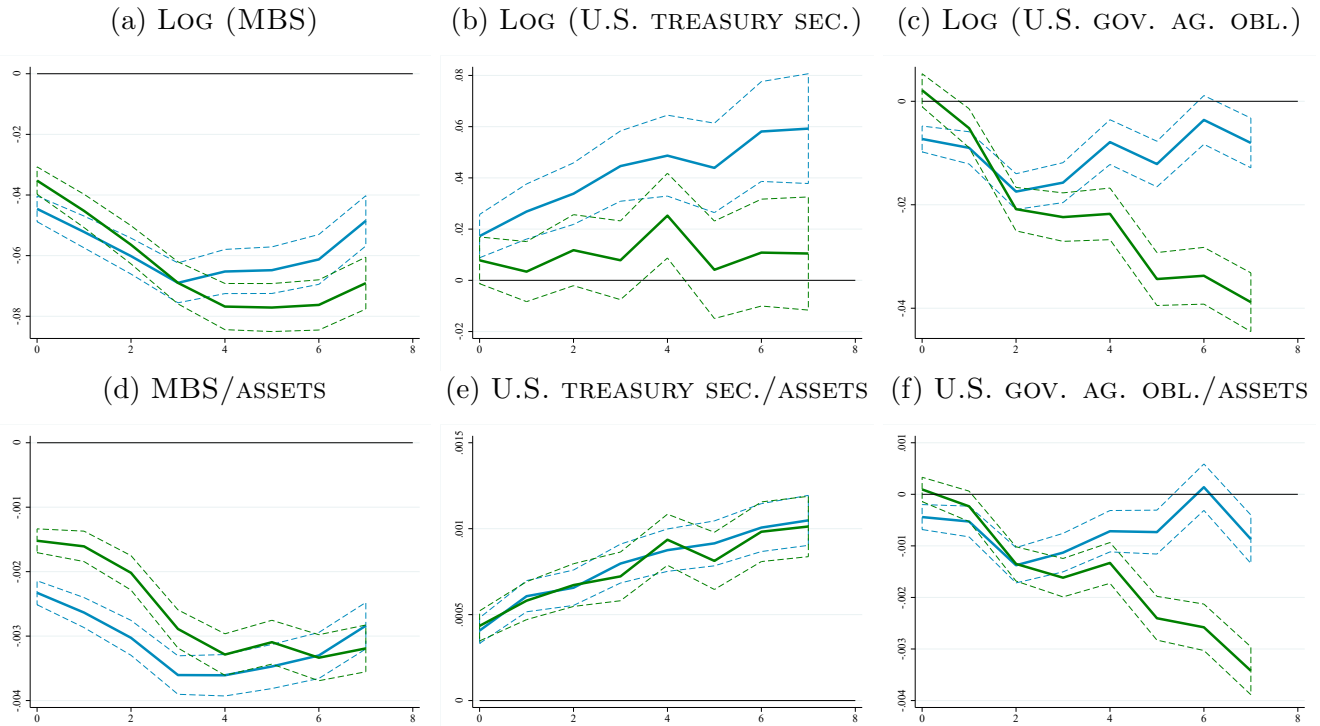
FIGURE A.18: LOCAL PROJECTIONS OF LOAN RATES TO MONETARY SHOCK BY DEPOSIT NII SHARE



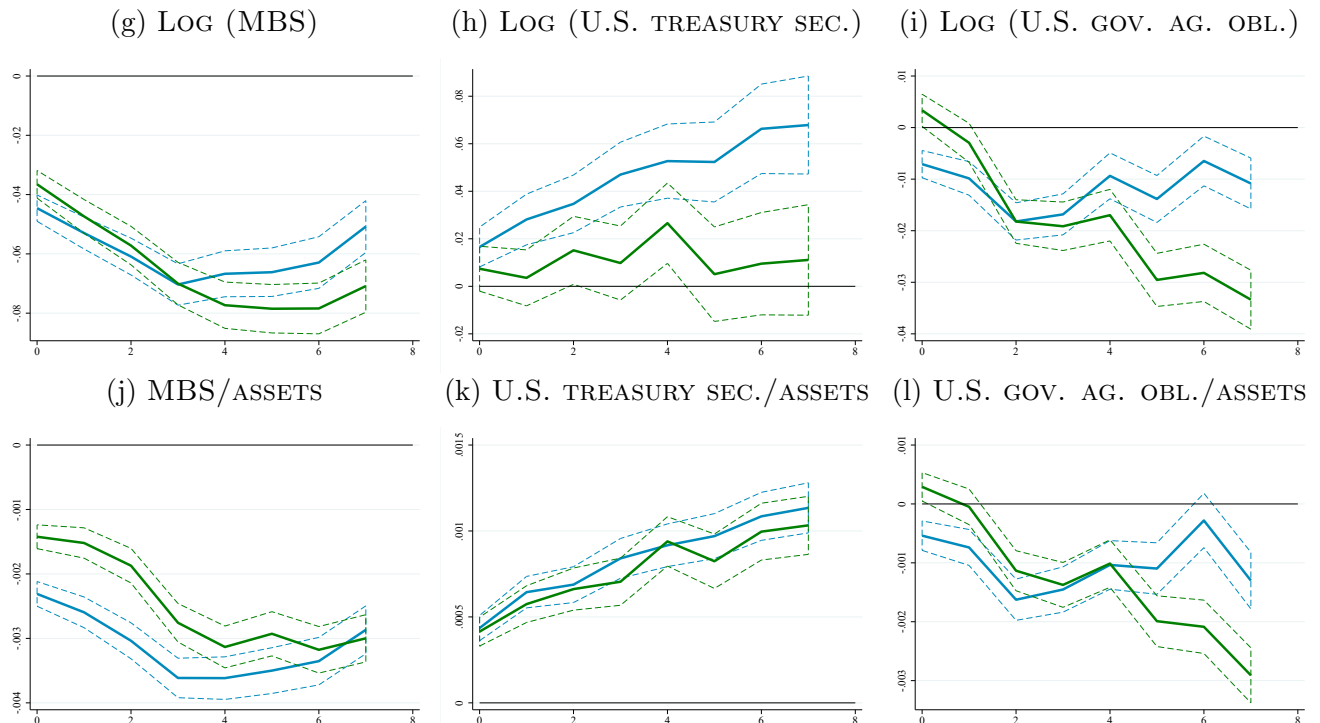
Notes: Impulse responses of the 1-year adjustable mortgage rate to a monetary policy shock at both high (blue) and low (green) shares of non-interest income. Horizon is in months. Standard errors are clustered by county. 90% confidence intervals. Source: RateWatch.

FIGURE A.19: BANK-LEVEL LOCAL PROJECTIONS BY DEPOSIT NON-INTEREST INCOME SHARE:
ADD. VARIABLES

Panel A: Without controls



Panel B: With size and capitalization controls



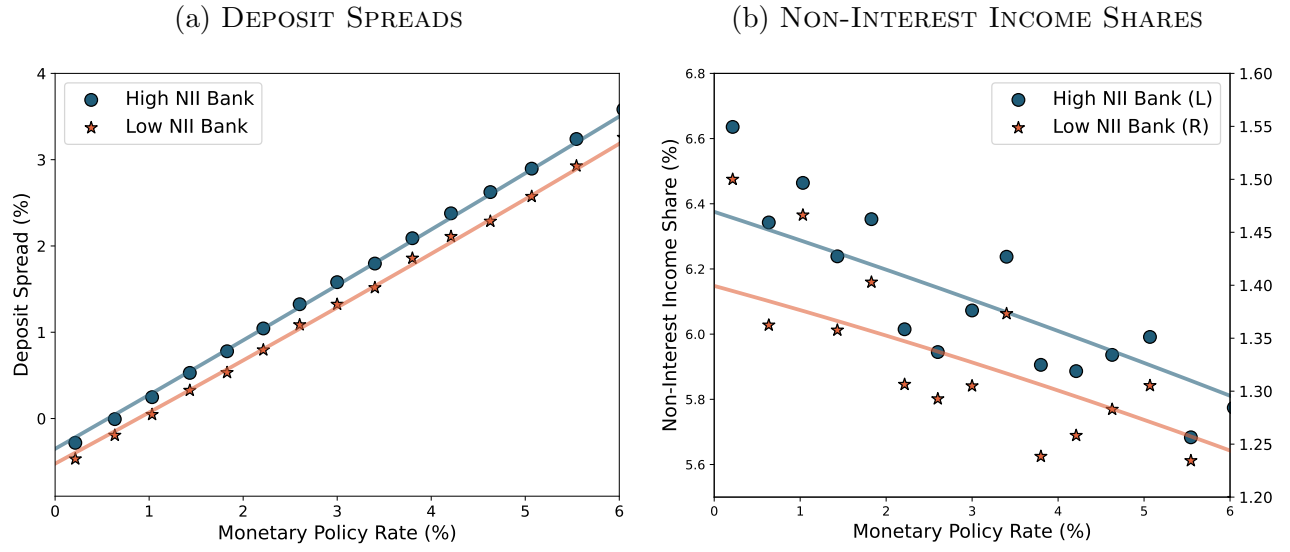
Notes: Impulse responses of mortgage-backed securities, U.S. treasury securities, U.S. government agency obligations in logs and as a ratio over total assets to a monetary policy shock at both high (blue) and low (green) shares of non-interest income. Panel B includes interactions of size and capitalization and the shock variable as controls. Horizon is in quarters. Standard errors are clustered by bank. 90% confidence intervals. Source: RateWatch.

TABLE A.2: UNTARGETED MOMENTS

Moment	Unit	Data	Model
Return on Assets _{high}	%	0.98	0.97
Deposit Rate Pass-Through _{high}	bps	[9,64]	43.8
Account Fee Pass-Through _{high}	%	0.20	0.78
Bank Z-Score _{high}	Ratio	22.1	17.7
Return on Assets _{low}	%	1.0	1.84
Deposit Rate Pass-Through _{low}	bps	[18,68]	53.0
Account Fee Pass-Through _{low}	%	0.29	0.40
Bank Z-Score _{low}	Ratio	23.0	31.7
Fee Difference	%	92.3	137.5

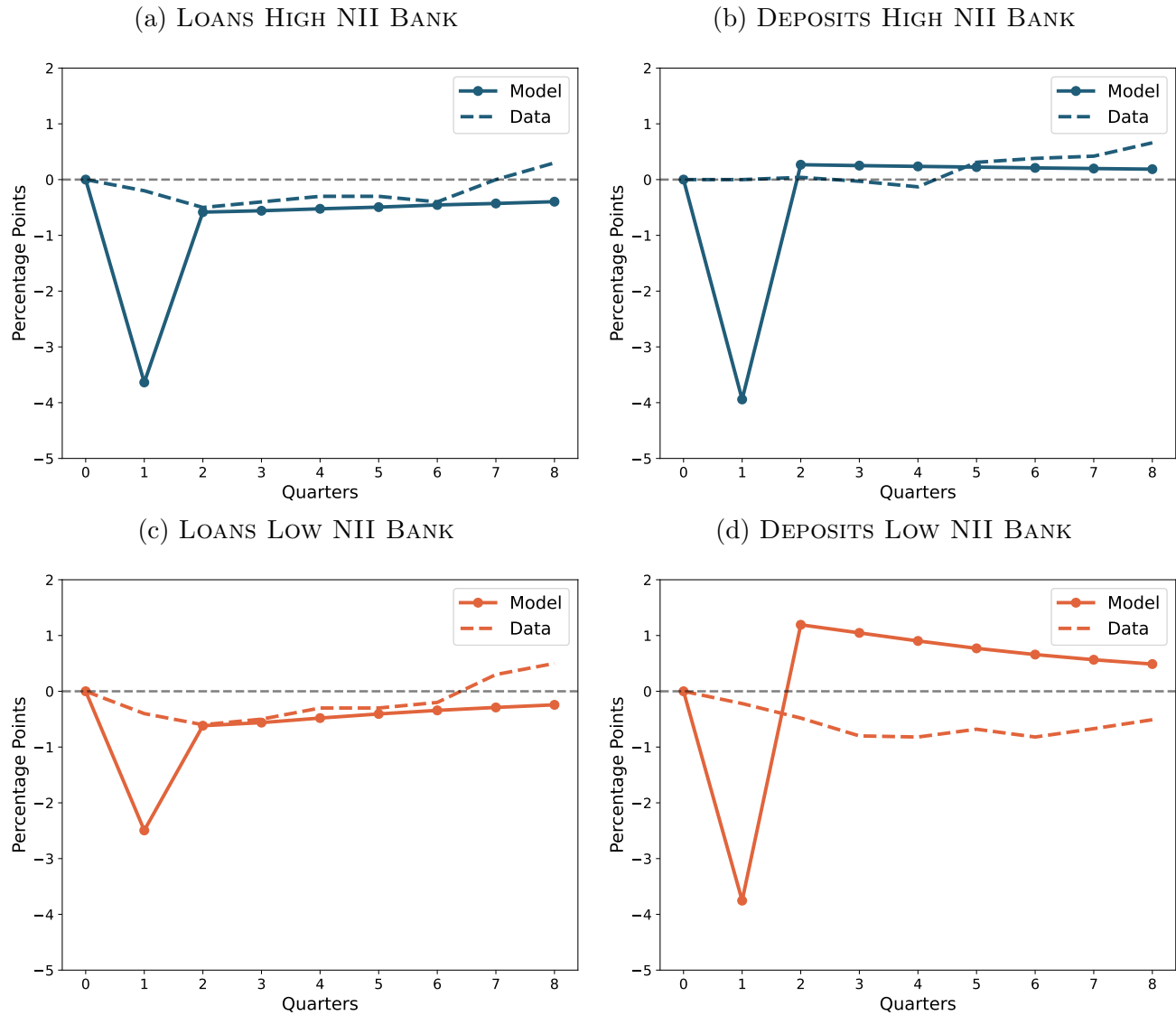
Notes: Model parameters are set for a model at a quarterly frequency. Data and Model moments are quoted in annualized terms. *High* reflects high deposit NII banks, and *low* reflects low deposit NII banks.

FIGURE A.20: BANK PRICING, INCOME AND MONETARY POLICY



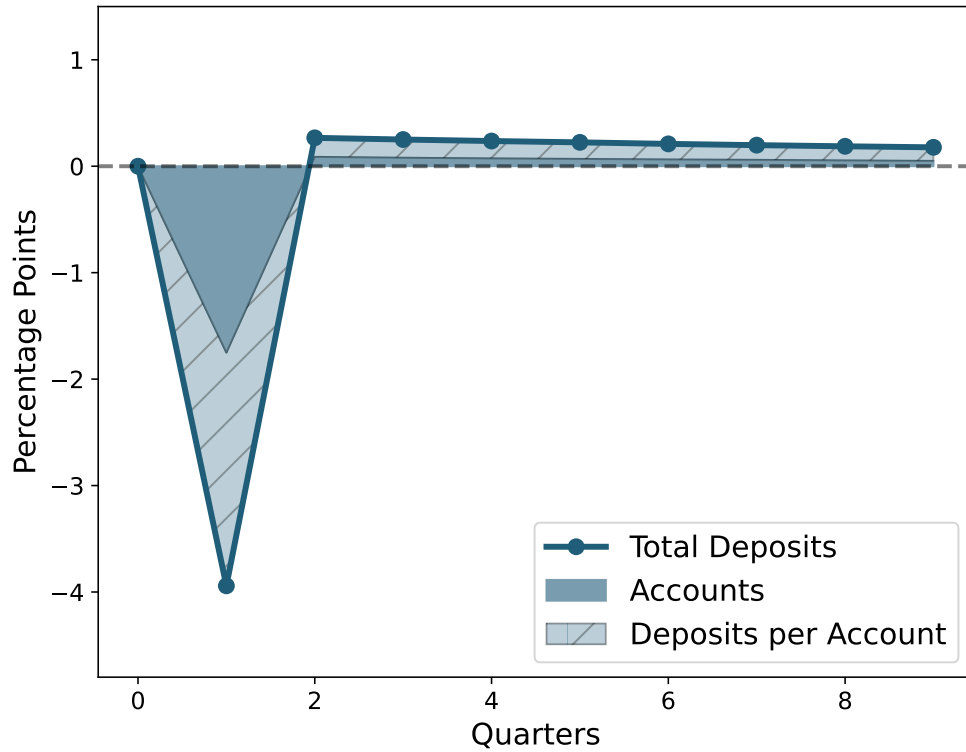
Notes: Equally spaced binned scatter collecting average moments for the high and low non-interest income bank. Fitted lines are included for illustrative purposes using a cubic interpolation.

FIGURE A.21: LOCAL PROJECTION RESPONSE FOR BANK QUANTITIES (MODEL AND EMPIRICAL MOMENTS)



Notes: Percentage deviation in loan and deposit quantities based upon empirical local projection estimates in and an internally-run model local projection in response to a 100 basis point monetary policy shock.

FIGURE A.22: EXTENSIVE AND INTENSIVE MARGIN RESPONSE TO MONETARY POLICY SHOCK



Notes: Percentage change deviation in total deposits for the High NII bank in response to a 100 basis point monetary policy shock. Changes in total deposits are decomposed into the extensive margin (change in *Accounts*) and the intensive margin (change in *Deposits per Account*).

TABLE A.3: BANK FEE CAP SETTINGS AND MODEL MOMENTS

Panel A: High NII Bank						
Fee Cap $\bar{f}_i$	100%	98%	96%	94%	92%	90%
Deposit Beta	0.35	0.35	0.36	0.36	0.36	0.36
Deposit Spread	103.66	66.30	56.35	49.42	44.42	40.09
Z score	17.62	-8.30	-12.26	-14.52	-15.92	-16.98
Lending	0.00	-15.02	-18.58	-20.93	-22.55	-23.92
Panel B: Low NII Bank						
Fee Cap $\bar{f}_i$	100%	98%	96%	94%	92%	90%
Deposit Beta	0.36	0.37	0.37	0.38	0.38	0.38
Deposit Spread	83.47	66.74	51.53	48.77	45.07	32.19
Z score	30.92	18.86	15.47	7.08	4.84	1.99
Lending	0.00	-4.39	-7.77	-9.19	-10.20	-13.90

Notes: This table reports average simulated moments from counterfactual versions of the model at various settings of the fee cap parameter $\bar{f}_i$. Fee cap values are expressed in terms of quantiles of the simulated fee distribution in the unconstrained, baseline model.

B Deposit Cost Estimation

We estimate a translog cost function similar to Berger, Klapper, and Turk-Ariss (2017), Berlin and Mester (2015), and Corbae and D’Erasmus (2021). For the estimation, we select a vector of outputs $\mathbf{q}_{it}$ and input prices $\mathbf{w}_{it}$ for bank i at time t to estimate the equation

$$\begin{aligned} \widetilde{\text{Cost}}_{it} = & \beta_1 \tilde{q}_{it}^\ell + \beta_2 \tilde{q}_{it}^d + \beta_3 \tilde{q}_{it}^{\ell 2} + \beta_4 \tilde{q}_{it}^{d 2} + \beta_5 \tilde{q}_{it}^\ell \tilde{q}_{it}^d + \sum_{j=1}^3 \gamma_j^\ell \tilde{w}_{ijt} \tilde{q}_{it}^\ell + \\ & \sum_{j=1}^3 \gamma_j^d \tilde{w}_{ijt} \tilde{q}_{it}^d + \sum_{j=1}^3 \sum_{k=1}^3 \phi_{jk} \tilde{w}_{ijt} \tilde{w}_{ikt} + \alpha_i + \xi_t + \epsilon_{it} \end{aligned} \quad (17)$$

where variables denoted with a tilde are log-transformed. We specify total lending q_{it}^ℓ and insured deposits q_{it}^d as the two bank outputs, and select three input prices related to the cost of labor, interest expense, and physical capital.³⁴ Our baseline specification uses net non-interest expense as the measure of Cost_{it} and *number* of deposit accounts as q_{it}^d , but also considers alternative specifications with gross non-interest expense and deposit volume, respectively. Refer to Section B for explicit details on variable selection and estimation. Once Equation (17) is estimated, a bank’s marginal cost of deposits can be expressed as

$$\widehat{mc}_{it}^{\text{deposit}} = \frac{\text{Cost}_{it}}{q_{it}^d} [\hat{\beta}_2 + 2\hat{\beta}_4 \tilde{q}_{it}^d + \hat{\beta}_5 \tilde{q}_{it}^\ell + \sum_{j=1}^3 \hat{\gamma}_j^d \tilde{w}_{ijt}] \quad (18)$$

³⁴We focus on insured deposits because they are a reasonable approximation of retail accounts which utilize/value depositor, and because we have available data on the *number of accounts*.

TABLE B.1: BANK TRANSLOG COST ESTIMATION

	Deposit Accounts		Deposit Volume	
	(1) $\log(\text{NIE}_{it})$	(2) $\log(\text{net NIE}_{it})$	(3) $\log(\text{NIE}_{it})$	(4) $\log(\text{net NIE}_{it})$
$\log(q_{it}^d)$	0.46***	0.130***	0.090***	-0.309***
$\log(q_{it}^d)^2$	0.003	0.014***	0.049***	0.074***
$\log(q_{it}^d)\log(q_{it}^\ell)$	-0.02***	-0.037***	-0.062***	-0.080***
$\log(\omega_1)\log(q_{it}^d)$	-0.042***	0.053***	-0.075***	-0.056***
$\log(\omega_2)\log(q_{it}^d)$	0.0**	-0.008***	-0.013***	0.018***
$\log(\omega_3)\log(q_{it}^d)$	-0.014***	-0.021***	-0.018***	-0.011***
Time + Bank FE	✓	✓	✓	✓
Other Controls	✓	✓	✓	✓
R^2	0.93	0.94	0.91	0.76
N	342,707	339,024	623,940	617,384

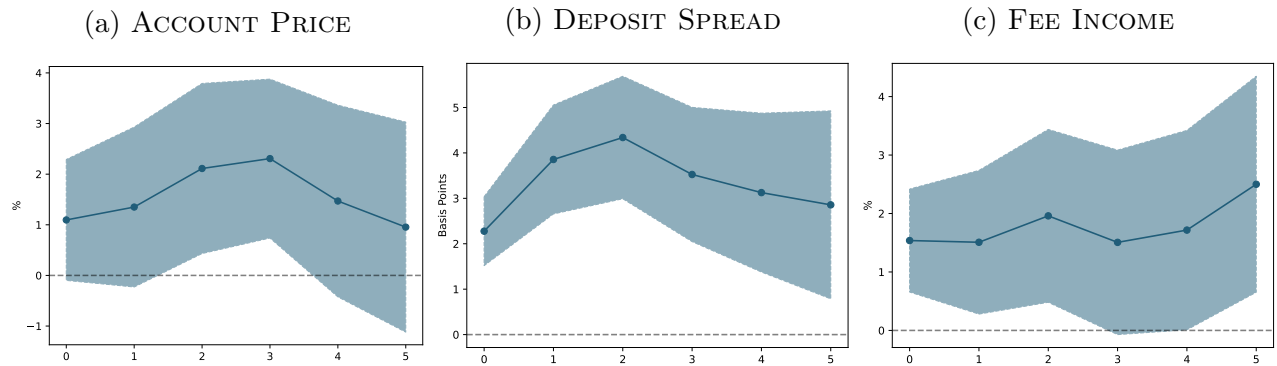
Notes: For columns (1) and (2), deposits q_{it}^d correspond to the number of insured deposit accounts for bank i in quarter t . For columns (3) and (4), q_{it}^d corresponds to the total volume of insured deposits for bank i in quarter t . The input prices $\{\omega_i\}_{i=1}^3$ correspond to the cost of labor, debt interest expense, and the cost of physical assets. All coefficients are estimated with robust standard errors. We denote the statistical significance of each estimate using stars (*p<0.1, **p<0.05, ***p<0.01).

TABLE B.2: HEDONIC COST REGRESSIONS

	(1)	(2)	(3)	(4)	(5)
	Net Cost / Assets (annualized bps)				
<i>Main Deposit Variables</i>					
Insured Deposits / Assets	0.967***	0.669***	0.574**	0.878***	0.791***
Uninsured Deposits / Assets	0.009	0.129	0.279**	0.128	0.280**
Small Time Deposits / Assets	-0.313	-0.415	-0.443	-1.669***	-1.712***
<i>High Non-Interest Income Interactions</i>					
High NII $\times$ Insured Deposits / Assets			0.410		0.401
High NII $\times$ Uninsured Deposits / Assets			-0.454***		-0.453***
<i>Bank Size Interactions</i>					
Insured Deposits / Assets $\times$ Size Q2				-0.162	-0.163
Insured Deposits / Assets $\times$ Size Q3				-0.352***	-0.364***
Insured Deposits / Assets $\times$ Size Q4				-0.173	-0.190
Small Time Deposits / Assets $\times$ Size Q2				0.847*	0.863*
Small Time Deposits / Assets $\times$ Size Q3				2.291***	2.303***
Small Time Deposits / Assets $\times$ Size Q4				1.767**	1.804**
<i>Control Variables</i>					
Loans / Assets		2.313***	2.334***	2.300**	2.319**
Foreign Deposits / Assets		-1.303	-1.250	-1.297	-1.242
Other Borrowed / Assets		-0.401	-0.357	-0.407	-0.363
Equity / Assets		2.162**	2.179**	2.130**	2.150**
Trading Assets / Assets		-1.868	-1.887	-1.678	-1.690
Trading Liabilities / Assets		-14.814***	-14.820***	-14.798***	-14.827***
Time FE	✓	✓	✓	✓	✓
Bank FE	✓	✓	✓	✓	✓
Bank Controls	X	✓	✓	✓	✓
R^2	0.004	0.030	0.031	0.032	0.033
N	15,003	15,003	15,003	15,003	15,003

Notes: The dependent variable is annualized net cost per assets (in basis points). All specifications include time and bank fixed effects. Robust standard errors are used. Statistical significance is denoted by: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Column (1) includes only main deposit variables. Column (2) adds balance sheet controls. Column (3) adds interactions with high non-interest income indicator. Column (4) adds bank size quartile interactions. Column (5) includes all variables. Similar to Drechsler, Savov, Schnabl, and Wang (2025), we restrict the sample to 2015-2021 and require that banks hold at least \$1 billion in assets as of December 2021, hold deposit-to-asset ratios larger than 65%, and drop banks with large variations between average quarterly assets and quarter-end assets.

FIGURE B.1: DEPOSIT PRICING AND PAST SPENDING ON DATA PROCESSING



Notes: We estimate a local projection model $y_{i,t+h} = \delta^h y_{i,t-1} + \beta^h \text{Data Share}_{it} + \xi_i + \gamma_t + \epsilon_{it}$ at the bank-year level where *Data Share* is the percentage of operating expenses attributed to data processing, normalized by its sample standard deviation. *Deposit spreads* are measured in basis points and all other dependent variables are log-transformed. The chart plots dynamic response β^h for each variable along with 95% confidence intervals, computed using robust standard errors.

C Illustrative Model Solution and Proofs

Consumers are distributed by taste parameter $\theta \sim U[0, \Theta]$ and maintain preferences over consumption and deposits $u(c, d) = c + v(d, \theta, r)$, where $v(d, \theta, r) = a(r)\theta d - \frac{d^2}{2}$. We assume $a(r)$ is an increasing function in r but do not provide an explicit functional form. Consumers receive an endowment ω and allocate the endowment between bonds b and deposits d such that $b + d = \omega$. In the case of opening a deposit account, consumption is determined by the return on savings net of fees such that $c = (1 + r)b + (1 + r^d)d - f$. These two constraints can be combined to express the consumer budget constraint as the budget constraint can be expressed as:

$$c = (1 + r)\omega - (sd + f), \quad (19)$$

where we define deposit spreads $s = r - r^d$. Substituting Equation (19) into the consumer utility function, the first-order condition for deposits can be written as $a(r)\theta - d - s = 0$ such that depositor demand is linear in spreads:

$$d^\theta(s, r) = a(r)\theta - s \quad (20)$$

and depends upon depositor-type θ and function $a(r)$, which represents a changing preference for liquidity. Without fees, depositor surplus can be expressed as $S^d(\theta, r, s) = v(d, \theta, r) - sd = \frac{(a(r)\theta - s)^2}{2}$. The problem of the bank choosing a fee is equivalent to choosing a threshold $\bar{\theta} \in [0, \Theta]$ such that the fee $f(\bar{\theta})$ fully extracts the surplus of type $\bar{\theta}$ depositors (i.e. $f(\bar{\theta}) = S^d(\bar{\theta}, r, s)$).

The bank's objective function is to maximize profits π via the profit function $\pi = (s - mc)D(s, f, r) + fN(s, f, r)$, where total deposit demand can be expressed as:

$$\begin{aligned} D &= \int_{\bar{\theta}}^{\Theta} d(s, \theta, r) dF(\theta) \\ &= \frac{6}{25} \frac{(a(r)\Theta - mc)^2}{a(r)\Theta} \end{aligned} \quad (21)$$

subject to the threshold $\bar{\theta}$ for participating consumers who open a deposit account. In addition, the total mass of participating depositors (i.e the number of deposit accounts) can be expressed as $N(s, f, r) = \frac{\Theta - \bar{\theta}}{\Theta}$. Then, the bank objective can be stated in terms of the tuple $(s, \bar{\theta})$ instead of

(s, f) . Specifically, the bank chooses a spread s and depositor threshold $\bar{\theta}$ to maximize

$$\begin{aligned}\pi &= (s - mc)D + fN \\ &= (s - mc) \frac{6}{25} \frac{(a(r)\Theta - mc)^2}{a(r)\Theta} + \frac{\Theta - \bar{\theta}}{\Theta} \frac{(a(r)\bar{\theta} - s)^2}{2}\end{aligned}\quad (22)$$

This now allows us to characterize the equilibrium conditions, as summarized in Proposition 1.

Proof of Proposition 1. The bank's first-order condition with respect to deposit spreads can be simplified to:

$$s^* + \frac{a(r)}{2}\bar{\theta}^* = \frac{a(r)}{2}\Theta + mc, \quad (23)$$

which implies $s^* = mc + \frac{a(r)}{2}(\Theta - \bar{\theta}^*)$. Further, the first-order conditions with respect to the depositor threshold can be written as:

$$3a(r)\bar{\theta}^* + s^* = 2a(r)\Theta + 2mc \quad (24)$$

which yields the solution:

$$s^* = \frac{4}{5}mc + \frac{a(r)}{5}\Theta \quad (25)$$

$$\bar{\theta}^* = \frac{3}{5}\Theta + \frac{2}{5}\frac{mc}{a(r)} \quad (26)$$

And using the surplus function $S^d(\theta, r, s)$, equilibrium fees can be stated as $f^* = \frac{2}{25}(A\Theta - mc)^2$. $\square$

Proof of Proposition 2. For part (i), define the price of a deposit account for type- θ depositors as $p^d(\theta) = f + sd^\theta(s, r)$. Then, take the derivative of the total price with respect to Θ ,

$$\begin{aligned}\frac{\partial p^d}{\partial \Theta} &= \frac{\partial f}{\partial \Theta} + \frac{\partial s}{\partial \Theta}d + s\frac{\partial d}{\partial \Theta} \\ &= \frac{\partial f}{\partial \Theta} + \frac{\partial s}{\partial \Theta}(d - s) \\ &= \frac{a(r)}{25} [2a(r)\Theta + 5a(r)\theta - 12mc]\end{aligned}\quad (27)$$

Thus, the sign of the derivative is positive if $2a(r)\Theta + 5a(r)\theta - 12mc > 0$. Using the equilibrium threshold $\bar{\theta}^* = \frac{3}{5}\Theta + \frac{2}{5}\frac{mc}{a(r)}$ for the marginal depositors, for which the constraint is most binding, the equation reduces to the condition $a(r)\Theta > 2mc$. Thus, given the regularity condition, $\frac{\partial p^d}{\partial \Theta}$ is

positive for all participating depositor types $\theta \in [\bar{\theta}^*, \Theta]$.

For part (ii), we observe $\frac{\partial s^*}{\partial \Theta} = \frac{a(r)}{5} > 0$ and $\frac{\partial f^*}{\partial \Theta} = a(r)\frac{4}{25}(A\Theta - mc) > 0$, given the assumed regularity condition that $a(r)\Theta > 2mc$.

For part (iii), the non-interest income ratio is expressed as $NII = \frac{fN}{fN+rD}$ where, in equilibrium, $Nf = \frac{4}{125} \frac{1}{a(r)\Theta} (a(r)\Theta - mc)^3$ and interest income can be expressed as $rD = r \frac{6}{25} \frac{(a(r)\Theta - mc)^2}{a(r)\Theta}$; thus, the ratio can be simplified to

$$NII = \frac{4(a(r)\Theta - mc)}{4(a(r)\Theta - mc) + 30r} \quad (28)$$

and the derivative $\frac{\partial NII}{\partial \Theta} = \frac{4a(r)}{Nf+rD} (1 - NII) > 0$ given NII is on the unit interval, total bank income is positive, and $a(r)$ is a non-negative valued function.

For part (iv), the derivative of spread with respect to the level of rates r is $\frac{\partial s}{\partial r} = \frac{a'(r)}{5} \Theta$ such that the cross-partial derivative with respect to market power is $\frac{\partial^2 s}{\partial r \partial \Theta} = \frac{a'(r)}{5} > 0$ given that A is an increasing function. $\square$

Proposition 3 provides a sufficient condition under which account fees have a stronger marginal effect on the participation decision of depositors to open an account.

Proposition 3. *Under equilibrium pricing, account fees exert a stronger marginal effect on depositor participation than deposit spreads whenever $a(r)\Theta - mc < 5/2$. This corresponds to the case in which the bank non-interest income share satisfies $NII^* < \frac{1}{1+3r}$.*

Proof of Proposition 3. The participation threshold is determined by:

$$S^d(\bar{\theta}, r, s) = f \quad (29)$$

where $\bar{\theta}$ is the marginal depositor. Using the surplus equation yields:

$$\bar{\theta}(s, f, r) = \frac{s + \sqrt{2f}}{a(r)}. \quad (30)$$

Fees have a stronger effect on the participation decision whenever $\frac{\partial \bar{\theta}}{\partial f} > \frac{\partial \bar{\theta}}{\partial s}$ which implies $f < \frac{1}{2}$. Using the equilibrium fee from Proposition 1, the condition $f^* < 1/2$ implies:

$$a(r)\Theta - mc < \frac{5}{2}. \quad (31)$$

Also, from Proposition 2, equilibrium non-interest income shares can be expressed as

$$NII^* = \frac{4(a(r)\Theta - mc)}{4(a(r)\Theta - mc) + 30r}. \quad (32)$$

and solving for $a(r)\Theta - mc$ gives:

$$a(r)\Theta - mc = \frac{30r}{4} \frac{NII^*}{1 - NII^*}. \quad (33)$$

such that substituting the condition from Equation (31) yields:

$$NII^* < \frac{1}{1 + 3r}. \quad (34)$$

□

Because the condition in Equation (31) lacks an empirical analogue, we utilize the condition on non-interest income shares in Equation (34), which can be directly observed in the data. For example, with a market rate of 2% (i.e. $r = 0.02$), the bank non-interest income share must be below 94% for fees to have a stronger effect on the participation decision than spreads. Since almost all banks in our sample operate with much lower non-interest income shares, we take this as supporting evidence that modeling account demand as a function of only fees in the quantitative model is a reasonable approximation.

D Model Equilibrium Conditions, Numerical Solution, and Counterfactual Exercises

Model Equilibrium Conditions. Define a bank's state as $\mathbf{s}_i = (n_i, z, r, i)$ such that a policy function $y(\mathbf{s}_i)$ represents a current-period decision and $y(\mathbf{s}'_i)$ represents a bank decision based upon the next period's state. Let $\lambda(\mathbf{s}_i)$ be the shadow multiplier on the bank's current-period budget constraint. The bank's first-order conditions can be stated as:

$$[r_i^d] : \frac{\partial d_i}{\partial r_i^d}(\mathbf{s}_i) - E[m(\mathbf{s}_i, \mathbf{s}'_i)] \left(r_i^d(\mathbf{s}_i) \frac{\partial d_i}{\partial r_i^d}(\mathbf{s}_i) + d_i(\mathbf{s}_i) \right) = 0 \quad (35)$$

$$[f_i] : \frac{\partial N_i}{\partial f_i}(\mathbf{s}_i) \left(f_i(\mathbf{s}_i) - \phi_i^n + d_i(\mathbf{s}_i) \right) - \frac{\partial N_i}{\partial f_i}(\mathbf{s}_i) E[m(\mathbf{s}_i, \mathbf{s}'_i)] (1 + r_i^d) d_i(\mathbf{s}_i) = 0 \quad (36)$$

$$[\ell_i] : - \left(1 + \phi_i^\ell \ell_i(\mathbf{s}_i) \right) + E \left[m(\mathbf{s}_i, \mathbf{s}'_i) \left(1 + r(z, z') + \Delta \right) \right] = 0 \quad (37)$$

$$[\pi_i] : 1 - \lambda(\mathbf{s}_i) (1 + \phi_i^\pi (\pi_i(\mathbf{s}_i) - \bar{\pi}_i)) = 0 \quad (38)$$

$$[BC] : \pi_i + \ell_i + \frac{\phi_i^\pi}{2} (\pi_i - \bar{\pi}_i)^2 + \frac{\phi_i^\ell}{2} \ell_i^2 = n_i + d_i N_i + N_i (f_i - \phi_i^n) \quad (39)$$

where $m(\mathbf{s}_i, \mathbf{s}'_i)$ is the bank stochastic discount factor, defined as $\beta \frac{E[\lambda(\mathbf{s}'_i)]}{\lambda(\mathbf{s}_i)}$ and $\lambda(\mathbf{s}_i)$ is determined by the dividend equilibrium condition. Then, for each point in the state space $\mathbf{s}_i$, solve for $\{r_i^d, \ell_i, f_i, \pi_i, \lambda_i\}$ using the system of equations (35)-(39).

Numerical Solution. We solve the bank's dynamic optimization problem using time iteration on a discretized state space. The state vector consists of bank networkth n_i , aggregate productivity z , the monetary policy rate r , and the bank's type i , which determines structural parameters affecting bank technology and demand. We construct a non-uniform grid for bank networkth with $N_n = 25$ grid points over the range with $n \in [-0.6, 12]$ using exponential spacing with growth rate 0.2 so that there is a larger concentration of grid points for low values of bank networkth. We discretize the AR(1) productivity process $z_{t+1} = \rho_z z_t + \sigma_z \epsilon_{t+1}$ where $\epsilon_t \sim N(0, 1)$ using the Tauchen method with $N_z = 6$ grid points spanning ± 3 standard deviations of the distribution. This yields a 6×6 transition matrix Π^z and grid points $z_{j=1}^6$. Last, we discretize the monetary policy shock using Gaussian quadrature (Miranda and Fackler 2002) with $N_r = 2$ nodes. The interest rate grid is determined by the support of the Taylor rule: $r_{\min} = \exp[\log(\bar{r}) + \phi_z \min(z_{j'} - z_j) + \min(\varepsilon^r)]$ and analogously for $r_{\max}$, with $N_r = 6$ evenly-spaced points between these bounds. The total state

space consists of $N_n \times N_z \times N_r \times |I| = 25 \times 6 \times 6 \times 2 = 1,800$ grid points.

We solve for equilibrium policy functions $\{\ell(\cdot), r^d(\cdot), \pi(\cdot), f(\cdot), \lambda(\cdot)\}$ using time iteration. Given a guess for the policy functions $\mathbf{y}^{(m)} = \{\ell^{(m)}, r^{d,(m)}, \pi^{(m)}, f^{(m)}, \lambda^{(m)}\}$ at iteration m , we,

Step 1: Compute expectation terms for each state (n_i, z_j, r_k) . Specifically, within equilibrium conditions (35)-(37), there are expectations over derivatives of the value function next period. With the use of the Envelope condition, these expectation terms can be stated in terms of the shadow multiplier $\lambda(\mathbf{s}')$ next period. So, for each state in the current period $\mathbf{s}_i$, the current guess for policy functions $\mathbf{y}^{(m)}$ is used to construct $\{E_{r^d}(\mathbf{s}_i), E_f(\mathbf{s}_i), E_\ell(\mathbf{s}_i)\}$ for the deposit rate, fee, and loan first-order conditions.

Step 2: Then, using the expectation terms from Step 1, solve the system of first-order conditions in equations (35)-(39) for each state (n_i, z_j, r_k) . We solve this system using the Levenberg-Marquardt algorithm. This yields updated solutions $\tilde{\mathbf{y}}^{(m+1)}$.

Step 3: Compute the mean squared error between the current solution and previous guess. If $\text{MSE}^{(m+1)} < 10^{-7}$ then exit loop. If the error threshold has not been reached, update the solution for the policy functions with dampening $\mathbf{y}^{(m+1)} = \delta \mathbf{y}^{(m)} + (1 - \delta) \tilde{\mathbf{y}}^{(m+1)}$, where we use $\delta = 0.3$.

To compute model moments, we simulate $T_{sim} = 10,000$ periods for each bank, recording outcomes for each period. When calculating model moments, we winsorize observations at the 1st and 99th percentiles to ensure robustness. For estimating model pass-through, we simulate each bank type over an 8-quarter horizon $T_{sim}^{shock} = 1,000$ times and measure impulse responses by comparing the baseline simulation with an otherwise identical simulation that includes a 100 basis point monetary policy shock in period 0. At each horizon, responses are measured as differences in average outcomes across the two simulations. These results are compared to a similar baseline simulation without the imposed 100 basis point monetary policy shock. We initialize bank networth at the baseline average for each bank type.

Last, the internal parameter $\bar{\pi}_i$ represents the long-run dividend target for each bank. The numerical solution outlined in Steps (1)-(3) takes the parameter $\bar{\pi}_i$ as given, but when solving the model, we must have an internally consistent value such that the simulated average $\hat{\pi}_i = \bar{\pi}_i$ for each bank type. Thus, for calibrating and solving the model, we introduce a fixed-point algorithm as an outer loop. In the outer loop, given a k^{th} guess $\bar{\pi}_i^{(k)}$, solve for policy functions via Steps (1)-(3) above, then simulate the model to obtain a sample average $\hat{\pi}_i^{(k)}$. Next, update the parameter value $\bar{\pi}_i^{k+1} = \frac{1}{2} [\bar{\pi}_i^k + \hat{\pi}_i^{(k)}]$ and repeat until $\max_i |\bar{\pi}_i^{(k+1)} - \bar{\pi}_i^{(k)}| < 10^{-4}$.

Exogenous Non-interest Income Counterfactual. For this counterfactual, we introduce a bank-level endowment $\bar{n}_i$ that augments the bank's budget constraint via:

$$\ell_i + \pi_i + C(\ell_i, \pi_i, N_i) = n_i + d_i N_i + f_i N_i + \bar{n}_i, \quad (40)$$

where $\bar{n}_i$ is an anticipated, persistent amount that augments a bank's funding. Importantly, for each value of $\bar{n}_i$, we re-solve the model and search for a new long-run dividend $\bar{\pi}_i$. For each bank, we consider levels of $\bar{n}_i$ relative to total net-interest income $(\text{net int})_i^{\text{base}}$ in the baseline model, such that a value of $p\%$ corresponds to $100 * \frac{\bar{n}_i}{(\text{net int})_i^{\text{base}}} = p$ in the counterfactual. Consequentially, a value of 0% corresponds to the baseline model in Table 8.

Fee Cap Counterfactual. In the fee cap counterfactual, we introduce an occasionally binding constraint:

$$f_i \leq \bar{f}_i, \quad (41)$$

to the bank problem, and solve for this by introducing a penalty function in the fee first-order condition $\chi^{\text{pen}} = \max\{0, f_i - \bar{f}_i\}^2$. Similar to the exogenous non-interest income counterfactual, we use simulated distributions from the baseline model to determine the level of the fee cap constraint—specifically, we set $\bar{f}_i^p$ based upon percentiles of bank i 's fees in the baseline model distribution of fees. So the value of $\bar{f}_i^p = p^{\text{th}}$ percentile of the baseline model's fee distribution. Consequently, in Figure 6, the value of 100% corresponds to the unconstrained, baseline model, and as $\bar{f}_i^p \rightarrow 0$ the fee cap becomes more restrictive.

E Data Definitions and Construction

Table E.1 reviews how bank statistics and measures are constructed. For most bank data, we rely on quarterly, merger-adjusted balance sheet and income data from bank Call Reports. The top two panels contain information for constructing measures of income and non-interest income in Table 1 and Table 2. The third panel contains information for constructing bank-level measures of deposit account price as in Figure 1, as well as bank measures for the summary statistics in Table 3. The final panel contains additional measures used for bank cost estimation related to the results in Table 4. We primarily use a sample period of 2000-2024, but also use 2010-2024 when using the *deposit account price* series which relies on observing the number of deposit accounts at the bank-level. Some data series are discontinued or experience a change in definition throughout the sample period. When this occurs, series are appropriately adjusted to maintain a consistent variable/measure. In addition, variables are winsorized at the 1st and 99th percentile to control for outliers, and all banks with less than 8 years of observations are dropped from the sample.

For the quantitative model, z-scores are constructed to match their empirical analogue:

$$z = \frac{\text{Capital Ratio} + \text{ROA}}{\sigma(\text{ROA})}, \quad (42)$$

where the capital ratio is defined as $(\frac{\ell - dN}{\ell})$, corresponding to book equity divided by total assets. Because deposits finance both lending and operating costs in the model, the capital ratio can become negative in low networth states when deposits exceed the stock of loans. Although the model does not explicitly incorporate insolvency or regulatory capital constraints, a negative capital ratio does not necessarily imply economic insolvency, as the bank's franchise value depends on the expected present value of future loan returns net of funding costs. Consequently, the model can generate negative z-scores. While such outcomes may appear counterintuitive from an accounting perspective, they arise naturally from the model's balance sheet structure and should be interpreted accordingly.

TABLE E.1: BANK VARIABLE CONSTRUCTION AND DEFINITIONS

Name	Definition	Call Report Item
Interest Income	total interest income	RIAD 4107
Total Income	total bank income	RIAD 4079 + 4107 + 14500
Non-Interest Income	total non-interest income	RIAD 4079
Operating Expense	total non-interest expense	RIAD 4135 + 4217 + 4092 + 4531
Deposit NII	deposit account fees	RIAD 4080 + C013 + C015 + C016
Non-Traditional NII	prop trading, investment banking, insurance, and fiduciary/consulting services	RIAD 4070 + 8757-8760 + F186
Asset NII	loan fees, securitization, and market gain/loss	RIAD B492 + B493 + 5416 + 5415 + B496
Other NII	residual	RIAD 4079 - (RIAD 4080 + C013 + C015 + C016) - (RIAD 4070 + 8757-8760 + F186) - (RIAD B492 + B493 + 5416 + 5415 + B496)
Deposit Beta	$\beta_{it}^d = \frac{r_{it}^d - r_{it-1}^d}{FFR_t - FFR_{t-1}}$ for savings accounts	N/A
Savings Deposit Rate	rate on savings accounts	$\frac{\text{RIAD 0093}}{\text{RCON 6810} + 0352}$
Deposit Spread	FFR minus savings rate	$\frac{\text{RIAD 0093}}{\text{RCON 6810} + 0352} \cdot \text{FFR}$
Number of Accounts	number of insured deposit accounts	RCON F046 + F050
Account Size	average insured deposit size at bank-quarter	$\frac{\text{RCON F045} + \text{F049} + 250 \cdot (\text{F048} + \text{F052})}{\text{Number of Accounts}}$
Deposit Account Price	total of deposit spread + deposit NII	See Equation (1)
Assets	total bank assets	RCON 2170
TCE Ratio	tangible common equity ratio	$\frac{\text{RCON G105} - (3164 + 3163 + \text{JF76})}{\text{RCON 2170} - (3164 + 3163 + \text{JF76})}$
RW Capital Ratio	total risk-weighted capital ratio	$\frac{\text{RCFA 3792}}{\text{RCOA A223}}$
Z-score	ratio of $\frac{\text{ROA} + \text{Capital Ratio}}{\text{ROA vol}}$	N/A
Deposit-to-Asset	total deposit to total asset ratio	$\frac{\text{RCON 2215} + 2385}{\text{RCON 2170}}$
Dividend Yield	dividend to asset ratio	$\frac{\text{RIAD 4475}}{\text{RCON 2170}}$
ROE	return on equity	$\frac{\text{RIAD 4340}}{\text{RCON G105}}$
ROA	return on assets	$\frac{\text{RIAD 4340}}{\text{RCON 2170}}$
Data Expense	bank expense on data processing	RIAD 0497
Marketing Expense	bank expense on marketing/advertising	RIAD C017
Non-interest Income Share	$\frac{\text{Non-Interest Income}}{\text{Total Income}}$	$\frac{\text{RIAD 4079}}{\text{RIAD 4079} + 4107 + 14500}$
Deposit NII Share	$\frac{\text{Deposit NII}}{\text{Total Income}}$	$\frac{\text{RIAD 4080} + \text{C013} + \text{C015} + \text{C016}}{\text{RIAD 4079} + 4107 + 14500}$
Net Non-Interest Expense	Operating Expense - Non-Interest Income	(RIAD 4135 + 4217 + 4092 + 4531) - RIAD 4079
Input Price ω_1	cost of labor	$\frac{\text{RIAD 4135}}{\text{RIAD 4150}}$
Input Price ω_2	cost of debt	$\frac{\text{RIAD 4172} + 4508 + 0093 + \text{HK03} + \text{HK04}}{\text{RCON 2215} + 2385}$
Input Price ω_3	cost of physical capital	$\frac{\text{RIAD 4217}}{\text{RCFD 2145}}$
Total Loans	total bank loans	RCON 5369 + B528 - 3123